

AI 時代光纖到晶片的新賽局

- 從矽光子、光互聯、CPO 到 AI 加速器的爭霸戰 -

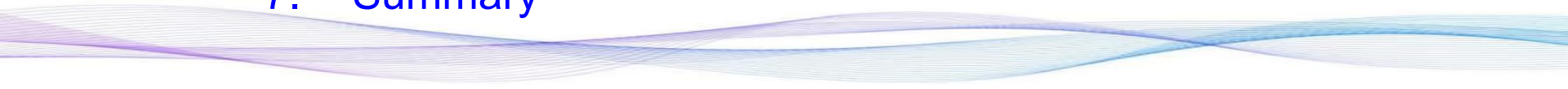
Foal Optics Corp. 小馬光學股份有限公司

Murphy Lin 林穎毅

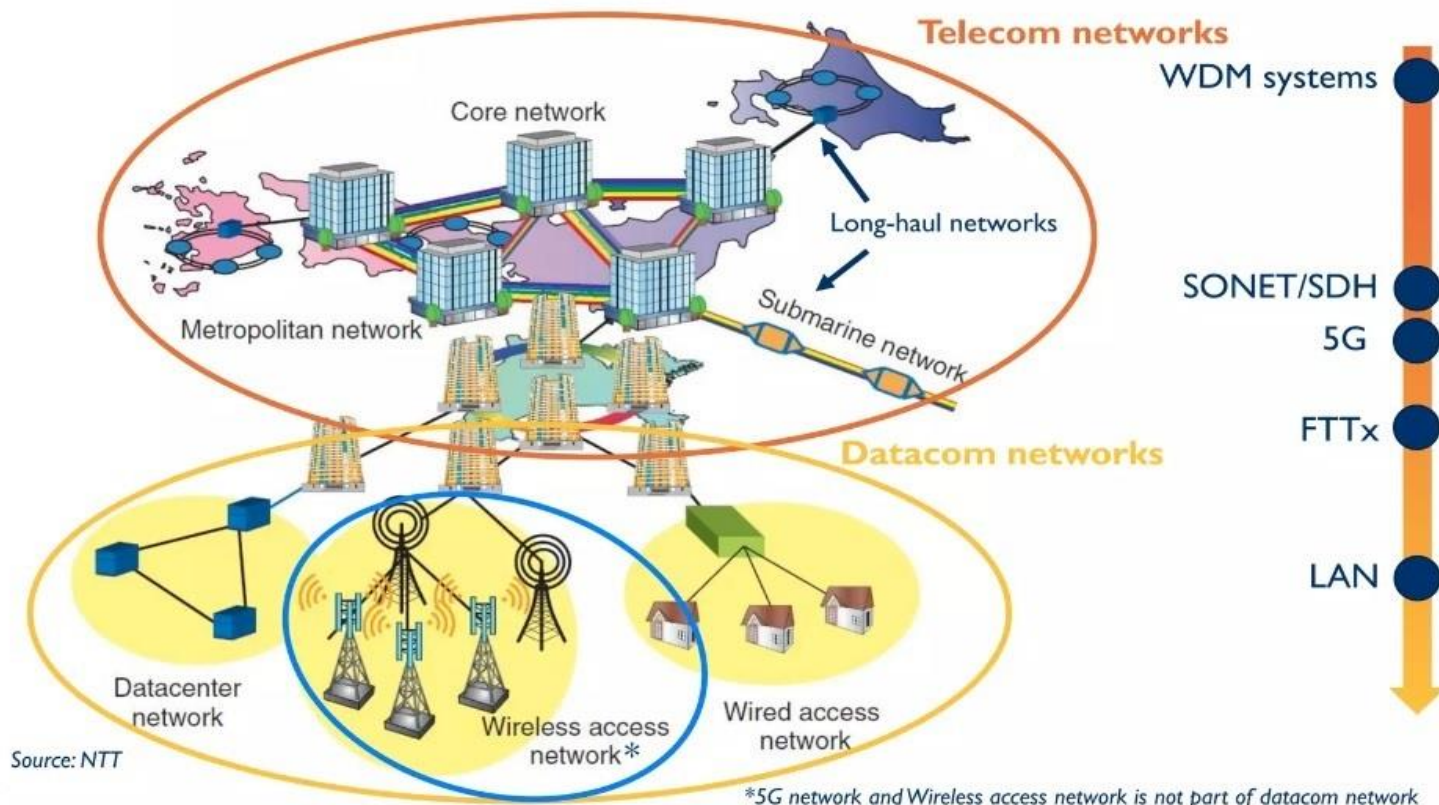
Chief Strategy Officer 策略長

Consultant, Taiwan Optics Association 台灣光學公會 顧問

演講內容

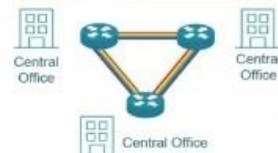
1. AI 算力需求如何帶動光通訊
 2. Interconnect 演進：Pluggable → LPO → XPO → NPO → CPO，與其市場區隔
 3. FAU & Coupling 挑戰與解決方案
 4. Silicon Photonics & Glass 成為 EPIC 平台
 5. Beyond Communication：光子技術的新戰場
 6. 全球矽光子產業可能的演進
 7. Summary
- 

各式不同的光纖網路



LONG-HAUL

- High performance
- Fiber constrained



METRO

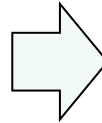
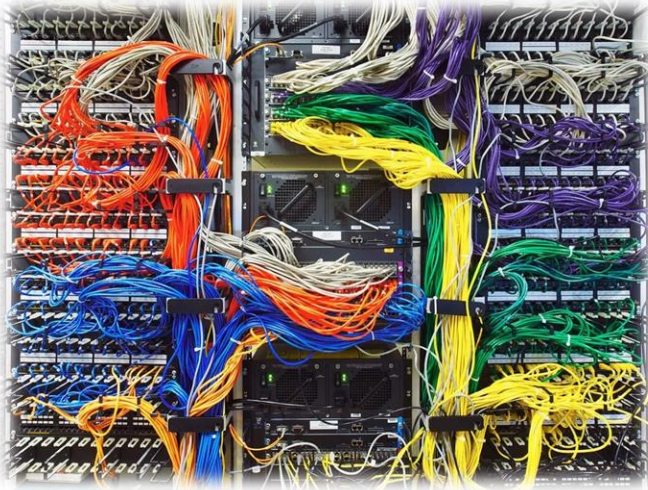
- Space and power constrained
- Pay as you grow model



EDGE

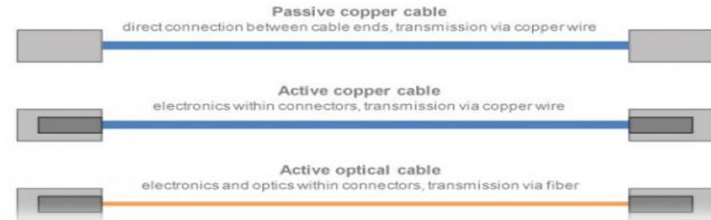
- Shorter product life cycles
- Cost and power sensitive

Copper or Fiber Cable in the Big Data Centers



	Passive copper	Active copper	Active optical
Cable span	Within racks (intra-rack)	Within and between racks (intra- and inter-rack)	
Cable attributes	Thicker, heavier cables with larger bend radii		Thinner, lighter cables with smaller bend radii
Cooling and airflow	Significant airflow blockage due to bulk of cable		Reduced airflow blockage
Power consumption	Distance-dependent	Fixed without regard to distance	
Electromagnetic interference (EMI)	Highly vendor-dependent		Fixed and low

Leading types of passive and active cables for the data center



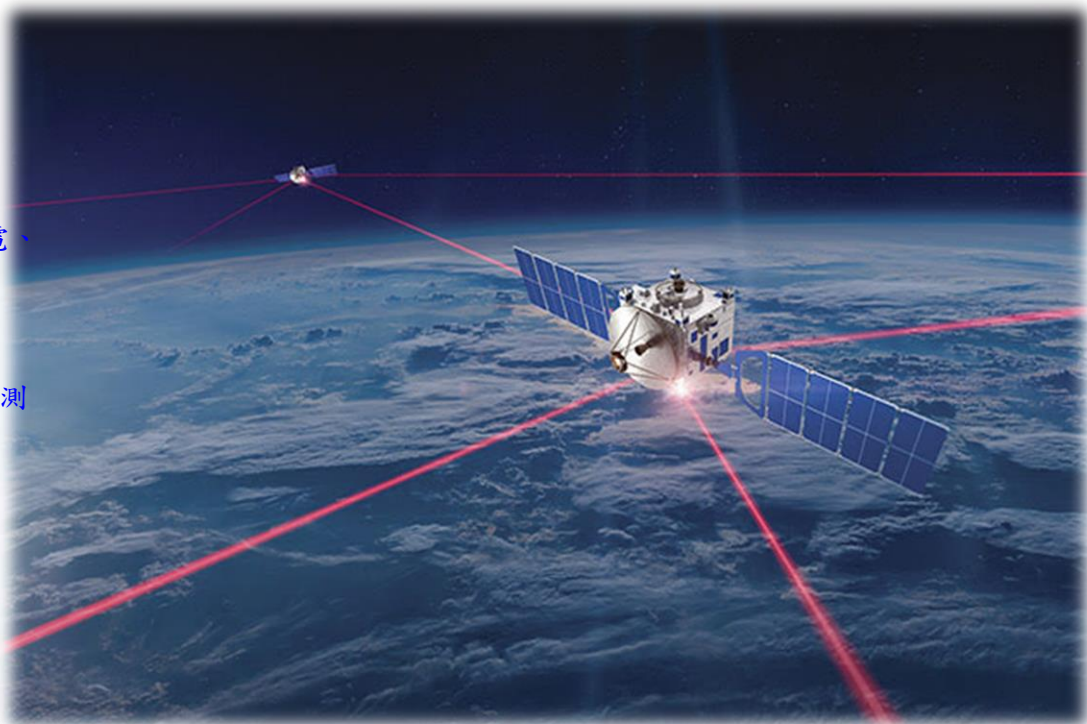
For the same data rate and application, optical cables (left) are often significantly lighter and less bulky than copper ones.

Major companies like Amazon, Microsoft, and Google are expected to spend over \$600 billion on infrastructure in 2026.

資料庫、光通訊上太空

衛星載體

1. 高增益天線
2. 各波段通訊模組：
 - C, Ku, Ka
3. 高效率太陽能光電、電池與電源系統
4. 酬載物件
 - ① 光學影像感測
 - ② 雷達
 - ③ 光譜儀



1. 合成孔徑雷達、雷射光通訊
2. 高速與高容量無線通訊
 - ① THz wave
 - ② All-optical network
3. 極低延遲與大量連結
4. 有線與無線網路控制技術
5. 非陸上多層無線系統
 - ① 航海通訊
 - ② 潛水艇通訊
6. 時空同步技術
 - ① 無線同步技術
 - ② 原子鐘
7. 資訊安全與可靠度
 - ① 量子加密
 - ② 電磁相容性
8. 實際創新應用
 - ① 遙測
 - ② IoT、自動駕駛
 - ③ 無人機應用

The global optical (laser) satellite communication market is experiencing rapid expansion, projected to grow from around \$0.6 billion in 2025 to over \$1.5 billion by 2030, with some projections suggesting a rise to over \$7 billion by 2034.

Integrated Photonics-Electronics Convergence System Technology

~2010

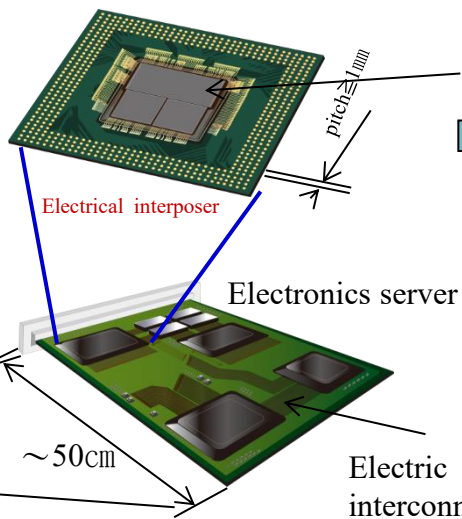
Server rack



10 server boards
in the rack

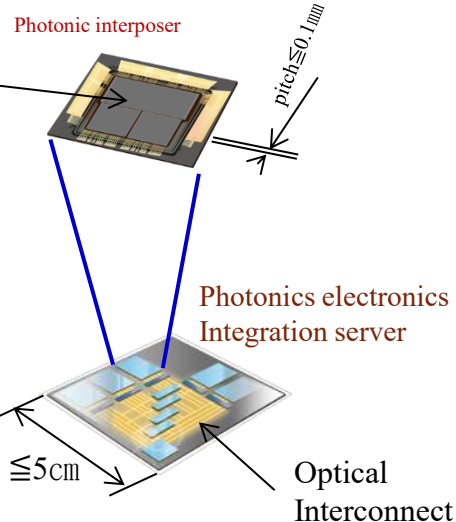
- ① Power consumption
- ② Interfacing area
- ③ Wiring density

Electronics



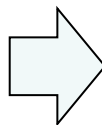
LSI

Photonics



2024~

Development targets



- 1/10: $\leq 1 \text{ mW/Gbps}$
- 1/100: $\leq 100 \text{ mm pitch optical connection}$
- x 10: $\leq 20 \text{ mm pitch optical waveguide}$

All-Photonics Network 全光網路 APN

全光子網路系利用矽光子技術以達到三個性能目標：低功耗、高品質和容量、更低延遲

Step 1

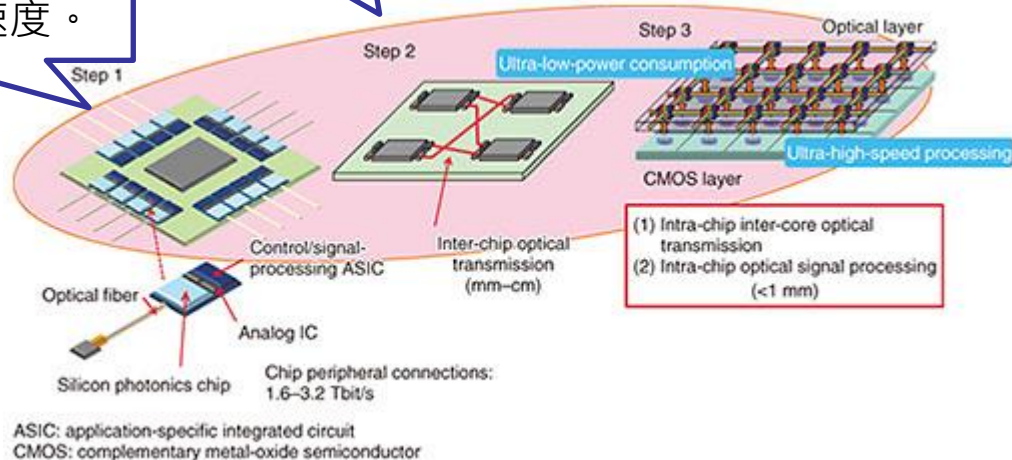
實現使用矽光子、光纖、類比積體電路 (IC) 等電路的整合，實現與晶片組外部元件的高連接速度。

Step 2

透過短距離光佈線直接互連晶片。

Step 3

透過光佈線在晶片內部進行核心間連接，實現超低功耗。



Step 3 - 仍須發展的技術：

- Optical Logic Circuit
- Optical Switch
- Optical Gateway
- Optical Transistor

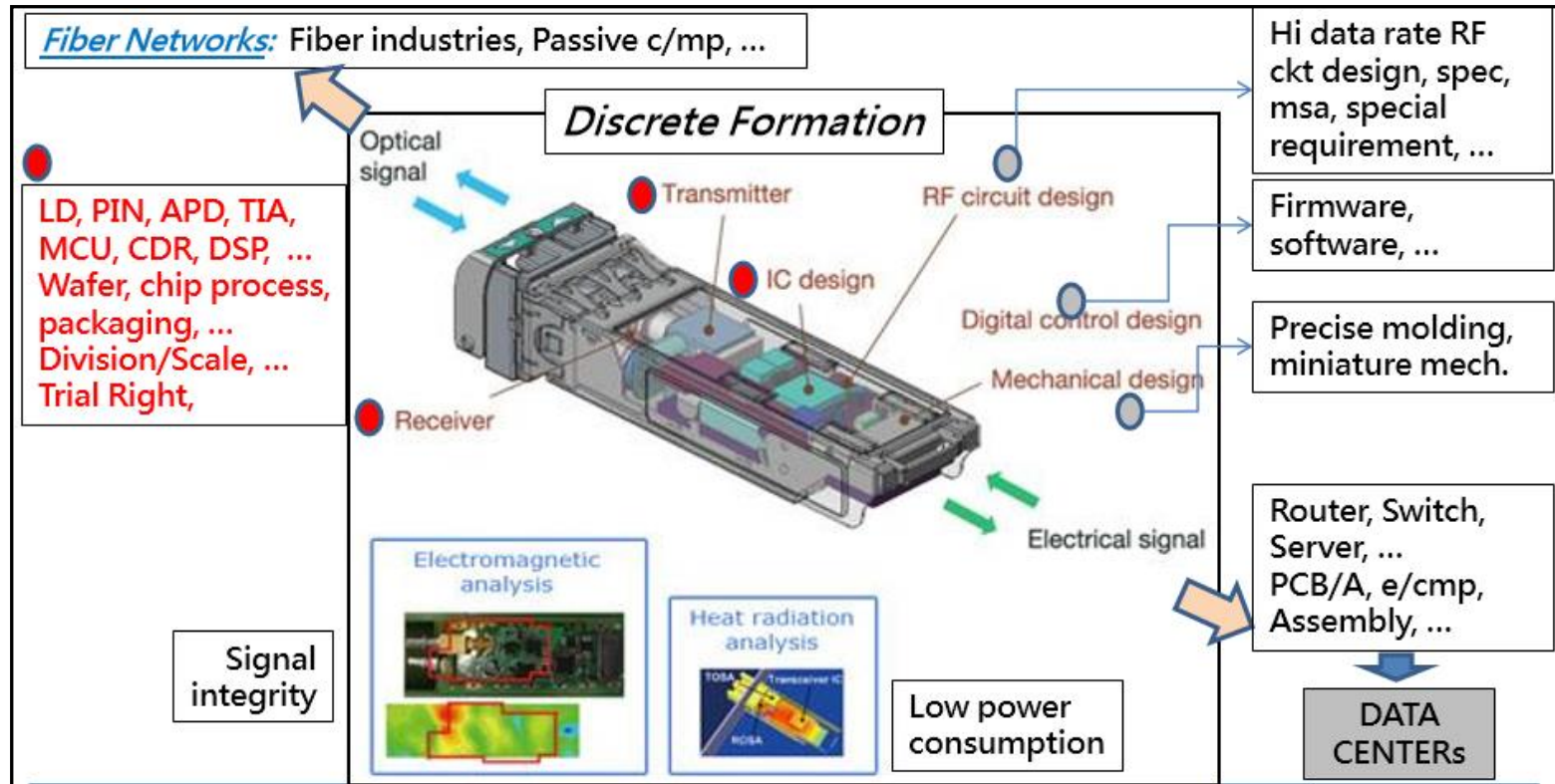
IOWN - Innovative Optical and Wireless Network

- APN: All-Photonics Network
- DTC: Digital Twin Computing
- Cognitive Foundation®

光收發模組 → 光引擎

Interconnect 演進：Pluggable → LPO → XPO → NPO → CPO，與其市場區隔

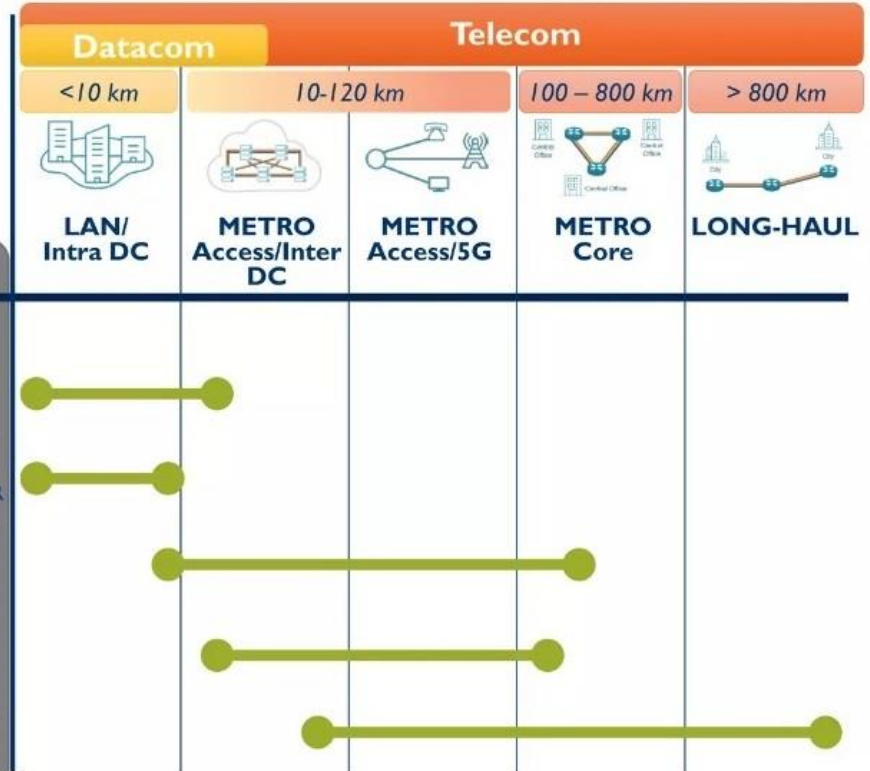
光收發模組：光源+光路+調變+光檢測



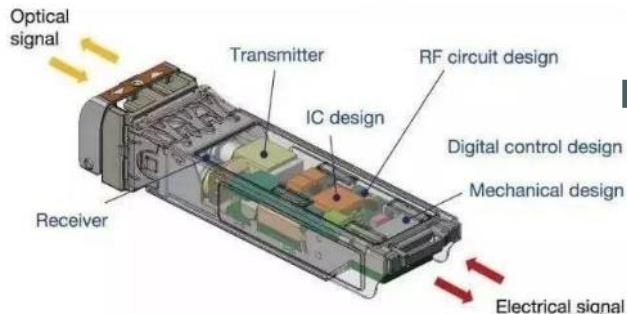
光源：Laser / 光被動元件：Passive: microlens, guide, filter, couple, split, combine / 光感測元件：photo detector

各式光收發模組在各通訊領域之應用

10G-25G	40G	100G	200G	400G
 SFP XFP SFP+		 SFP28		
 QSFP	QSFP+	QSFP28 – SR/DR/FR/ CWDM/PSM	QSFP56 – SR/DR/FR	QSFP-DD OSFP – SR/DR/FR
		QSFP28– LR/ER/LWDM	QSFP56 – LR/ER	QSFP-DD OSFP – LR/ZR
	 CFP2-LR/ER	 CFP4-LR/ER		 CFP8-LR
		5"x7" module 4"x5" module CFP-DCO	CFP2-ACO CFP2-DCO	OSFP/QSFP-DD ZR/ZR+



Transceiver → Optical Engine

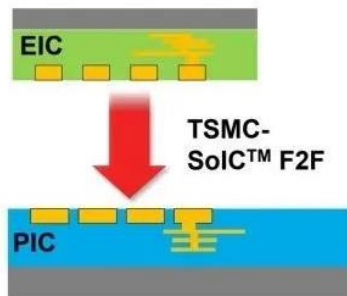


tsmc COUPE - Compact Universal Photonic Engine
緊湊型通用光子引擎

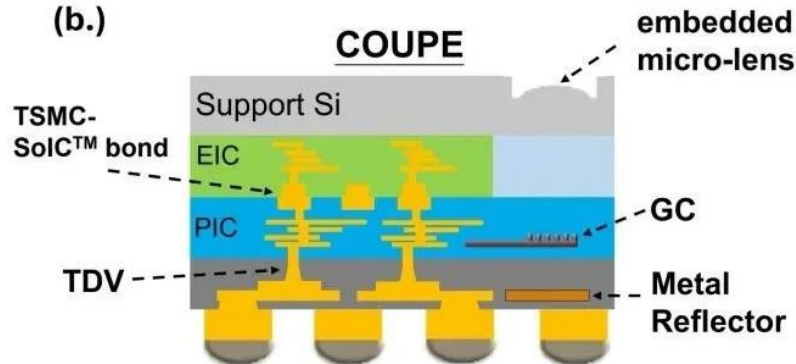
台積電COUPE光引擎

項目	完成驗證時間	型式	速度
第一代 COUPE	2025年	可插拔式光模組	1.6 Tbps
第二代 COUPE	2026年	整合CoWoS封裝成CPO	較第一代減少兩倍功耗 跟十倍訊號延遲
第三代 COUPE	未 訂	結合2.5D封裝，將光引擎、XPU與HBM晶片封裝在矽中介板上	預計再減少五倍功耗 跟兩倍訊號延遲

(a.)



(b.)

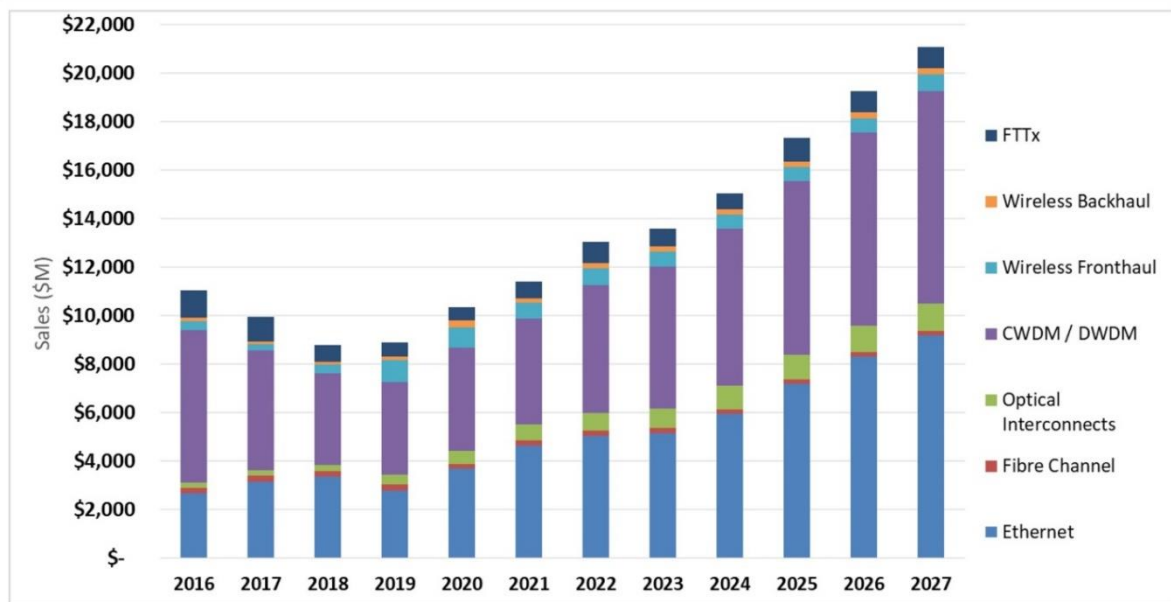


- (a) TSMC-SolC™ face-to-face (F2F) technology for EIC and PIC bonding.
- (b) COUPE critical components consist of TSMC-SolC™ bond, TDV, embedded micro-lens, and metal reflector.
- (c) TDV - Trench Dielectric Via 介質通孔

光收發模組市場有數百多億美元 2025~2035 CAGR 9.8%

Vendor

Global sales of optical transceivers 2016-2027 by segment



Source: LightCounting

- II-VI
- Acacia
- Accelink
- Applied Optoelectronics
- ATOP
- Avago (FOIT)
- Broadcom
- Cisco
- Coherent
- ColorChip
- Delta (台達)
- Eoptolink (新易盛)
- Finisar
- Fujikura
- Furukawa
- HG-Genuine
- Hisense (海信)
- Hitachi Cable
- Innolight (中際旭創)
- GigaLight
- Kaia
- Lumentum
- Luxtera
- Methode
- Molex
- NEC
- NeoPhotonics (新飛通光電)
- Oclaro
- OE Solutions
- Oplink (光聯通訊)
- Samtec
- Source Photonics (索爾思光電)
- SuperXon
- Sumitomo
- TE Connectivity

Optical Communication for Interconnect

各式光互連之技術路線

矽光子 - 操縱“光”的整合平台

- 光的維度：波長 / 頻率、振幅、偏振、方向、波前、相位、同調、色散、軌道角動量、自旋、量子態...

- 光源：LASER, Micro LED

- 決定光的波段
- 連續 / 非連續
- 同調性、強度

- 幾何光學：鏡片、菱鏡

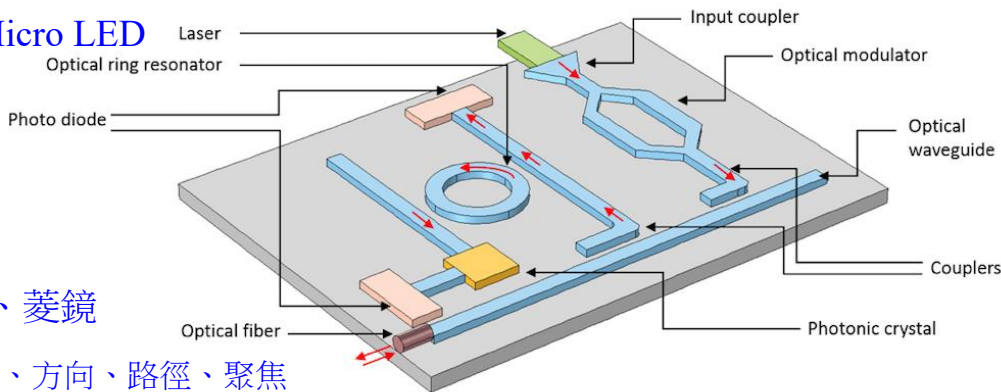
- 控制光的傳播、方向、路徑、聚焦

- 波動光學：濾鏡、光柵、DOE、SLM

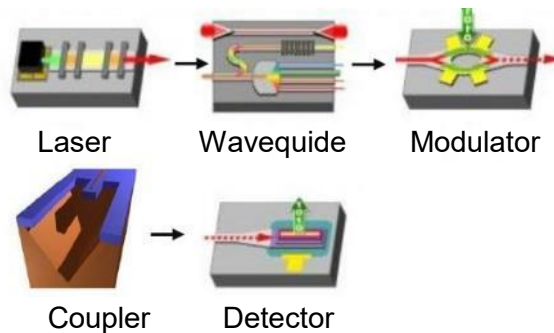
- 控制光的波段、相位、波前、干涉、繞射

- 量子光學

- 操縱光的疊加態、糾纏態



- Laser
 - 光源 (1310~1550nm)
- Coupler 耦合器
 - 增加入射光量
- Modulator
 - 調變光的相位
- Waveguide & Fiber
 - 控制光的傳播
- Photo Diode
 - 光的偵測
- Resonator 共振
 - 增加特定波段的振幅
- Photonics Crystal
 - 波長轉換



矽光整合

SILICON PHOTONIC INTEGRATION

SILICON PHOTONICS INTEGRATED CIRCUITS

ACTIVES

Hybrid laser
Electrons generate photons



SiGe Photodetector
Photons generate electrons



Si Modulator
Electrons control photons

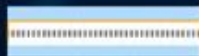


PASSIVES

MUX/ Demux



Optical Filter



Splitter/Combiner



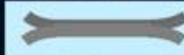
Coupling I/O



Interferometer/Switch



Polarization Diversity

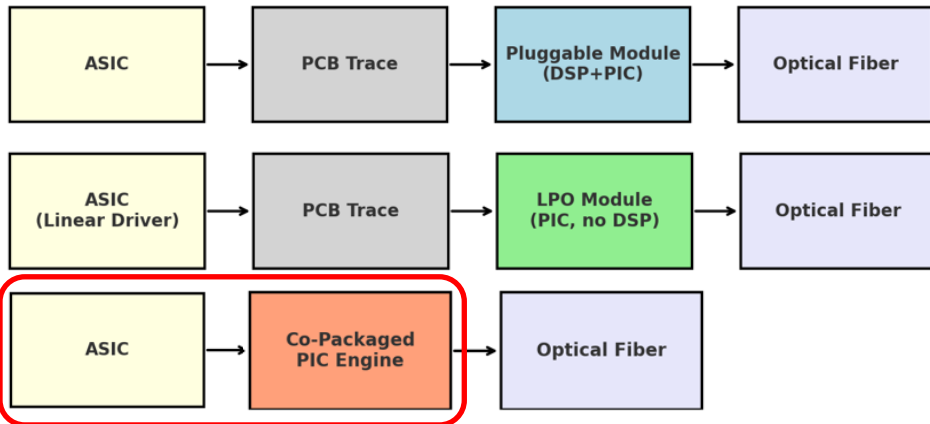
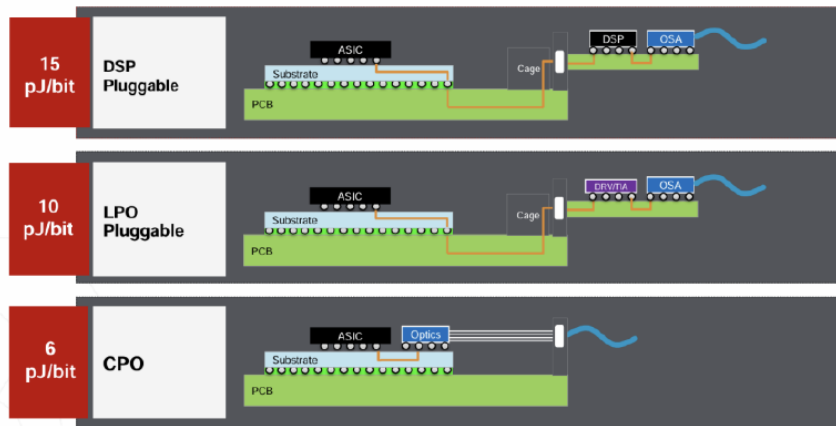


WAVEGUIDE INTEGRATION

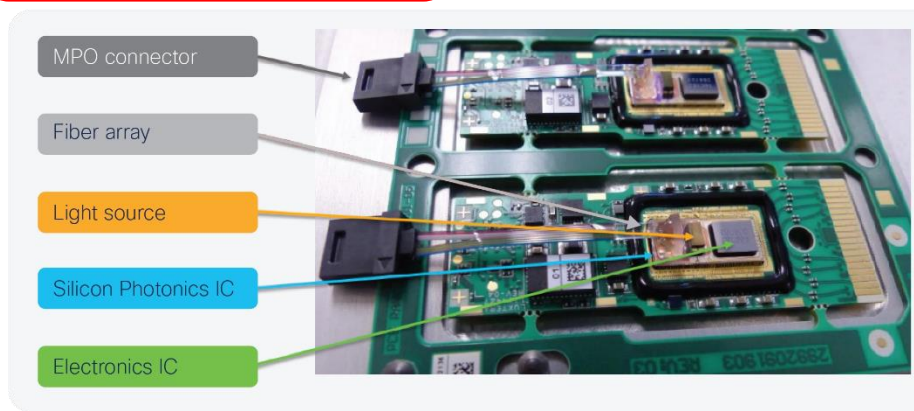
Large-scale PIC &
On-chip Interconnects



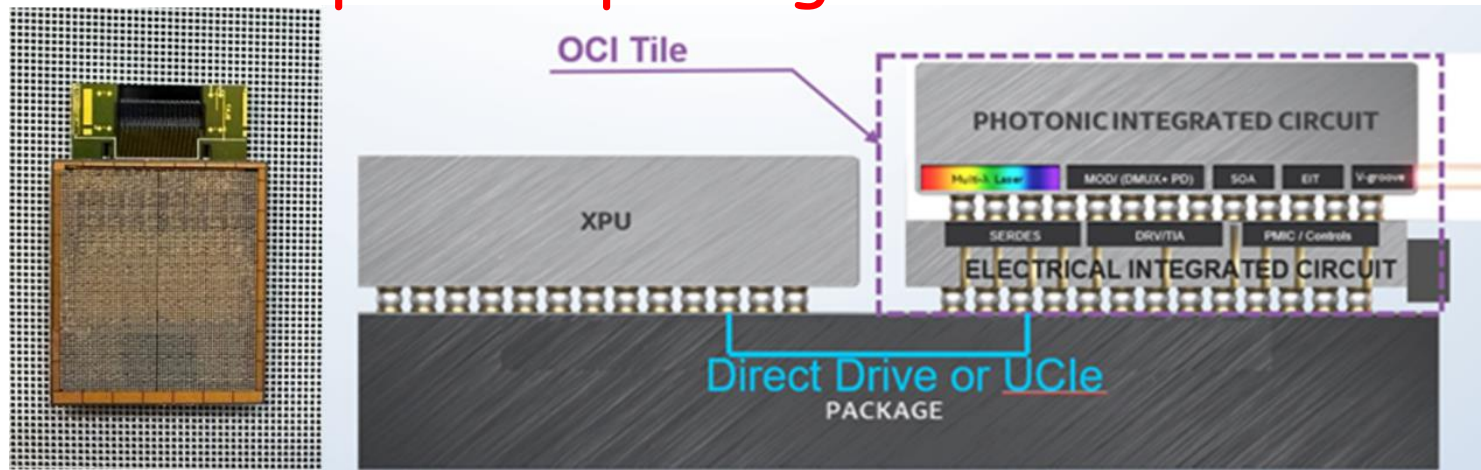
Architecture Evolution of Transceiver -> LPO -> CPO



Feature	Pluggable	LPO	CPO
Power Efficiency	~15–20 pJ/bit	~8–12 pJ/bit	<5 pJ/bit (target)
Latency	High (DSP overhead)	Low (No DSP)	Very Low (Direct)
Bandwidth Density	Limited (front panel)	Limited	Very High (chip-edge I/O)
Maintainability	Hot-swappable, easy to service	Hot-swappable	Low, requires board replacement
Maturity	Very mature	Emerging	Early stage

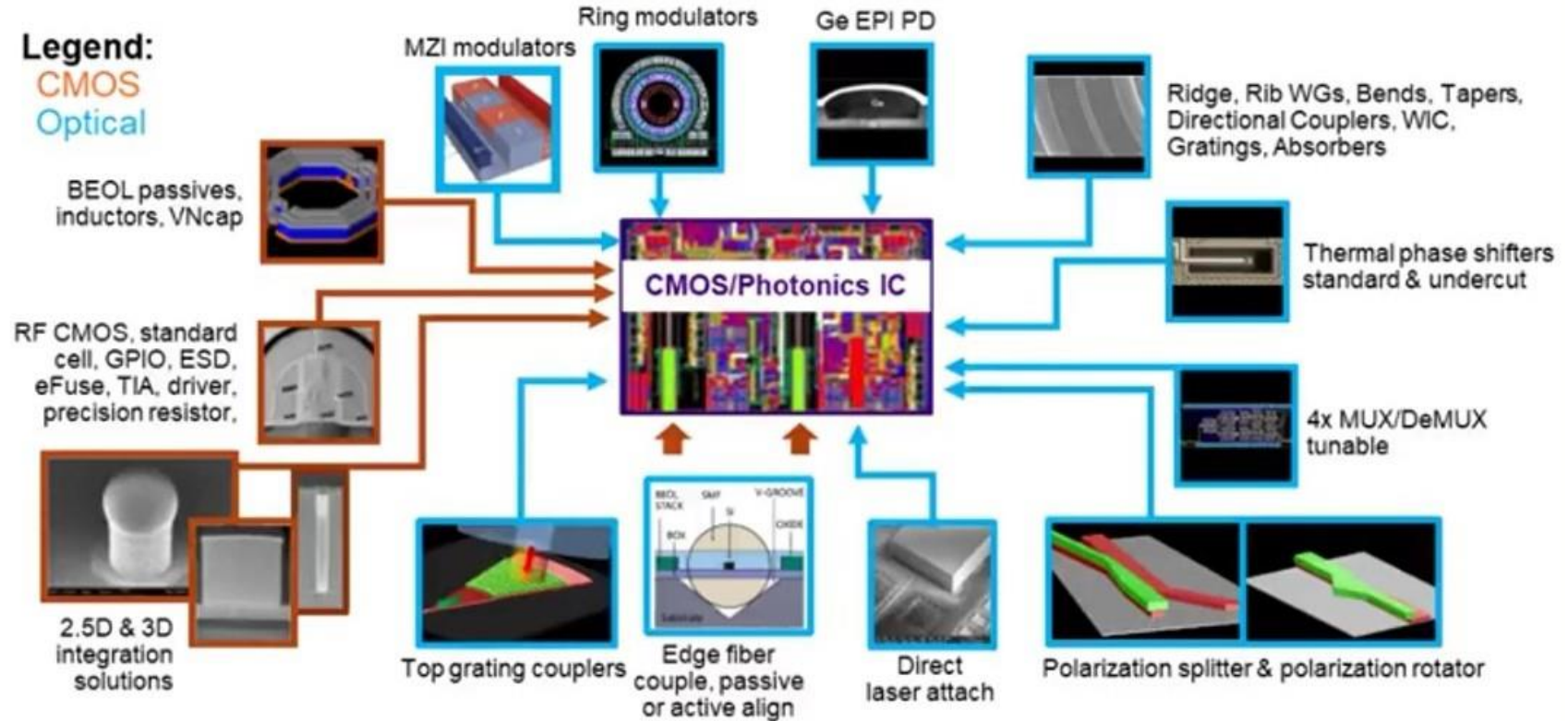


Intel: Optical Compute Interconnect (OCI) chiplet co-packaged with XPU



- Intel 的OCI旨在透過在CPU、GPU和加速器之間建立高頻寬、低延遲和低功耗的光數據通道，克服傳統電氣互連的瓶頸。
- Intel 採用晶片組架構，利用先進封裝將不同的功能模組（CPU、GPU、PIC、I/O、記憶體控制器等）整合到單一封裝中。
- 每個光通道可支援數百Gbps的傳輸速率，並且透過並行多通道波分多工（WDM）技術，可擴展至20~40 Tbps等級。
- 光收發器不再獨立存在，而是與CPU/ASIC封裝在一起，即Integrated Laser + SOA + PIC，無需外接雷射光源。
- 已累積 800 萬顆以上 Silicon Photonics PIC 出貨經驗。
- 從 Switch-CPO 邁向 Compute Optical I/O，為下一階段矽光子的重要演進方向。

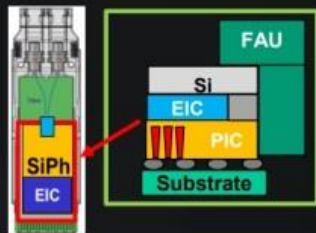
GlobalFoundries: Fotonix platform



3D Optical Engine (OE) for Next-Gen Communication

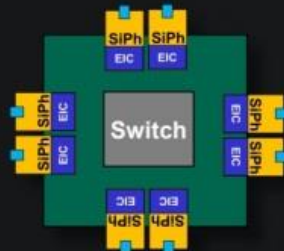
- Optics is crucial for rapid and reliable data transmission and lowering network power consumption for AI
- EIC-on-PIC stacked using the SoIC-X process (COUPE™*) offers unparalleled interconnect density while maintaining optimal system power
- We will enable COUPE in pluggable in '25, followed by COUPE on substrate in a CoWoS CPO with a 2x reduction in power and a 10x reduction in latency in '26
- COUPE on CoWoS interposer for another 5x reduction in power and 2x reduction in latency is being explored

- Pluggable (OSFP**)
- 1.6Tbps OE on Substrate in 2025



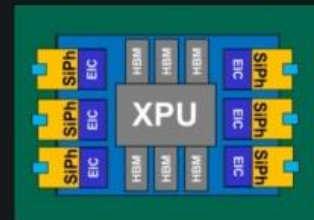
Power: 1X
Latency: 1X

- CPO with Switch
- 6.4Tbps OE on Substrate in 2026



< 0.5X
< 0.1X

- CPO with XPU
- 12.8Tbps OE on Interposer in Pathfinding



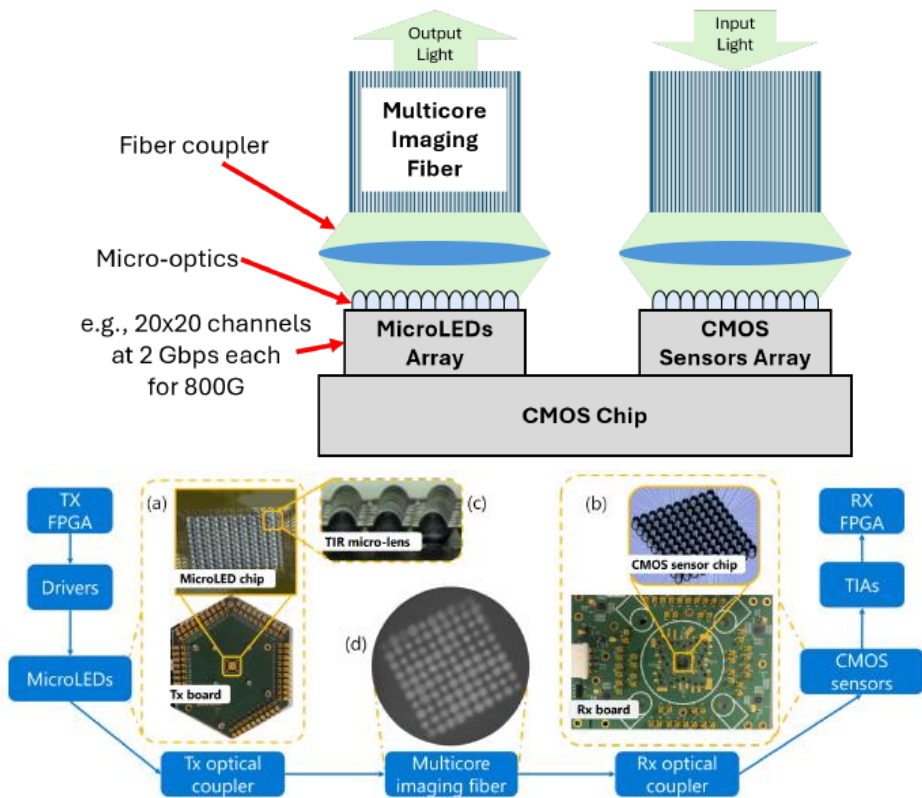
< 0.1X
< 0.05X

* COUPE: Compact Universal Photonic Engine

** OSFP: Octal Small Form Factor Pluggable

Microsoft: MOSAIC- Micro-Optical Synthesis and Integration of Components

MOSAIC's high-level WaS architecture and key components.



1. 打破傳輸距離與功耗的權衡

- 銅纜鏈路高效可靠，但傳輸距離通常小於 2 公尺。
- 光纖鏈路則能擴展傳輸距離。

2. 寬而慢 (Wide-and Slow, WaS) 架構

- MOSAIC 架構並非採用少數高速通道（窄而快），而是使用數百個低速並行光通道。
- 每個通道的運行速度約為 2 Gbps，可輕鬆擴展至 800 Gbps 以上的總容量。

3. 核心技術 – Micro LED + Imaging Fiber + Analog Backend

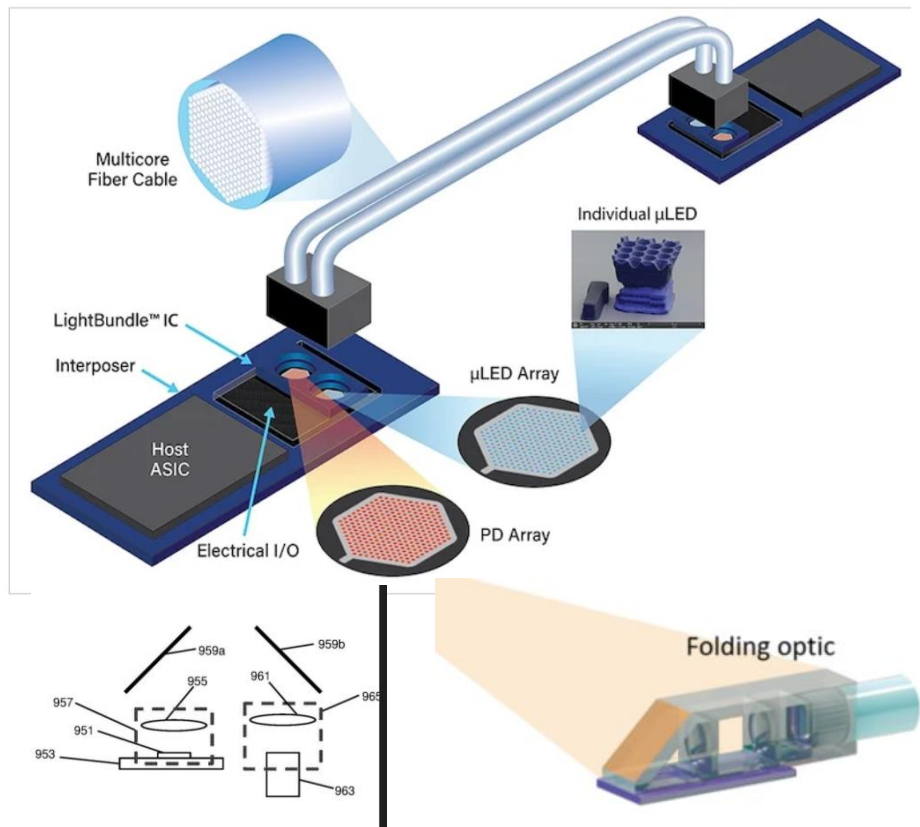
- 微型 LED：超低功耗，直接調製
- 多芯成像光纖：一根光纖中包含數千個纖芯，承載多個通道

4. 資料中心和 AI 應用

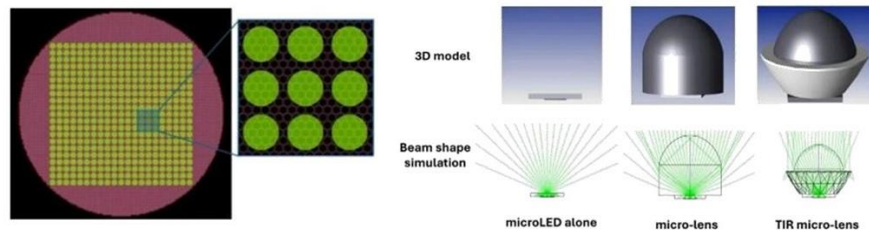
- 可直接替代 QSFP/OSFP 封裝。
- 協定無關（相容於乙太網路、PCIe、CXL、NVLink）。
- 開啟了新的設計空間：解耦記憶體、更大的 GPU 叢集、簡化的拓撲結構。

(a) transmitter and (b) receiver PCB with fabricated Micro LEDs and CMOS sensors (insets). (c) Fabricated **TIR (total internal Reflection)** micro-lens array. (d) 100-channel image at the output of the imaging fiber.

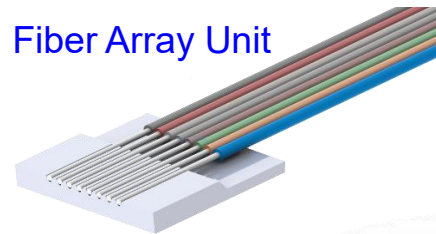
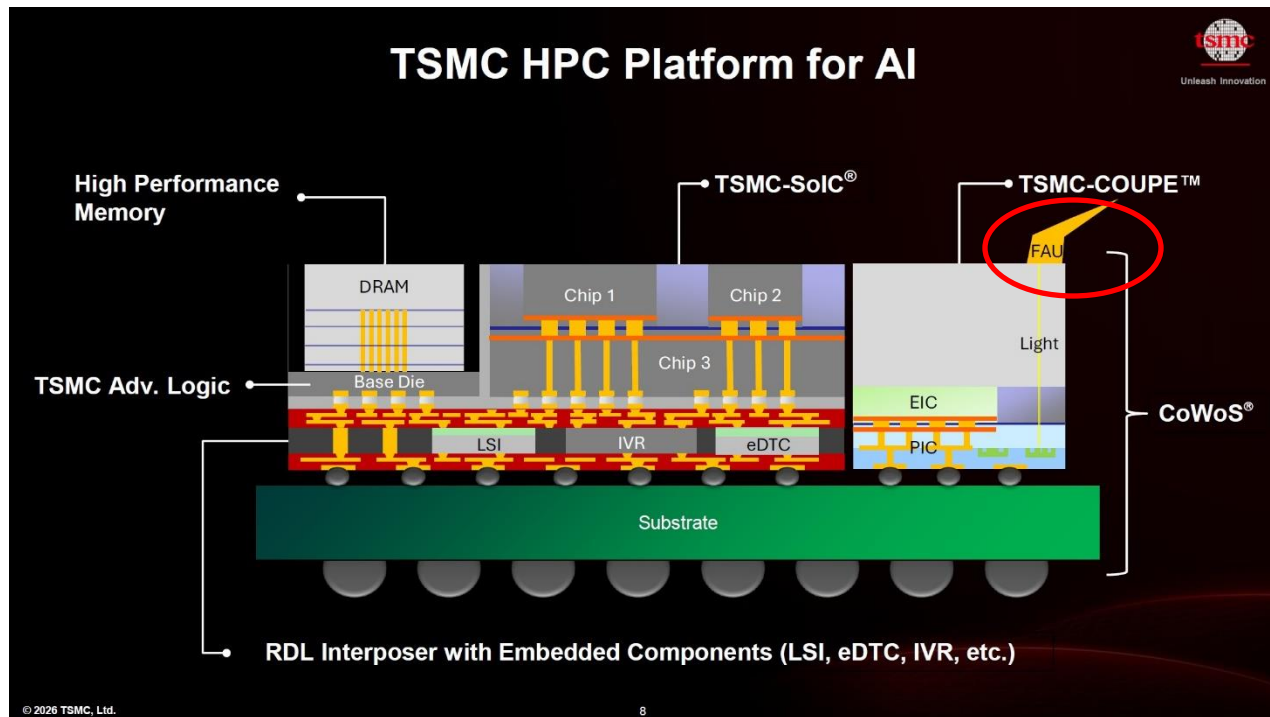
Avicena: Micro Led, PD, Fiber bundle的耦合



- 基於MicroLED的光互連
 - 採用GaN Micro LED實現晶片間和模組間互連。
- Terabit-Class 頻寬密度
 - LightBundle™：200根光纖陣列，間距 $30\mu\text{m}$ ，耦合至光電偵測器陣列。每個通道的資料速率高達10 Gbit/s。
 - 在幾公尺範圍內實現 $10\text{Tbit}/\text{mm}^2$ 的頻寬密度。
- 低功耗
 - 與傳統的SerDes或基於雷射的光互連相比，其功率效率是一項主要優勢。
 - 功率效率為 $0.1\text{pJ}/\text{bit}$
- 應用
 - 應用在資料中心裡的儲存系統、AI加速器互連和共封裝光學元件（CPO）。



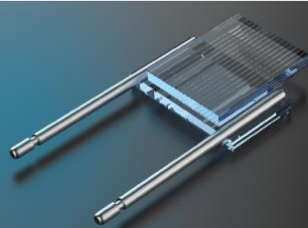
tsmc's COUPE™-with FAU



Challenges:

- 精密機械之精細度如何看齊半導體業
- 公差匹配
- 光耦合效率
- 幾何光學 → 波動光學
- 良率、插拔損失
- 通道數增加？

康寧 GlassBridge™ Fiber-to-PIC Connector 概要



CORNING

GlassBridge™ Fiber-to-PIC Connector

Industry-leading design for next-generation optics

Built on Corning's decades of expertise in glass science, ion-exchange (IOX) waveguides, and industry-proven TMT-ferrule technology, the GlassBridge™ connector gives system designers a practical, manufacturable path to next-generation optical architectures including Near-Packaged Optics (NPO), Co-Packaged Optics (CPO), and high-density photonic modules.

Advanced Optical Coupling

- Wafer-based high-volume manufacturing
- Customizable pitch
- Detachable solution
- Passive alignment

Inside the GlassBridge Design – Core Product Features

 Wafer-based IOX waveguides enabling high-volume, scalable production, and precision alignment.	 Supports 24 fibers and can scale to multiple (e.g. 2 x 24) per PIC for higher density needs.	 Robust physical-contact interface leveraging a standard TMT ferrule format.
 Fiber-to-chip coupling using an ultra-compact, 6.4 mm-wide glass connector body.	 Solder reflow compatible to ease assembly.	 Low loss optical performance demonstrating 1.5 dB O-band fiber-to-PIC coupling.

Corning Optical Communications

GlassBridge™ Fiber-to-PIC Connector | OCM-152-ADN | Page 1

• 產品特點 Advanced Optical Coupling

- Wafer-based waveguides：能實現產量，以及精確的對準。
- Supports 24 fibers：可以擴展到多個（例如，2 x 24 個）的配置。
- Fiber-to-chip coupling：6.4 mm 寬的玻璃連接器。
- Passive alignment：有效的組裝方式可以節省成本。
- Low loss optical performance：1.5 dB O-band fiber-to-PIC coupling.

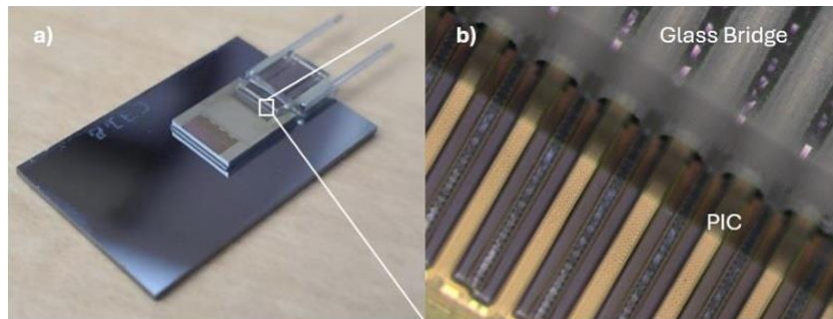
• 系統級優勢

- Customizable pitch：支持多種PIC間距，如40, 27, 165 μm 。
- Detachable Connector
- 標準化 TMT-ferrule：利用標準 TMT-ferrule 接口以確保可靠性和易用性。
- 支持定制系統架構：可整合多個 GlassBridge 解決方案以滿足特定需求。

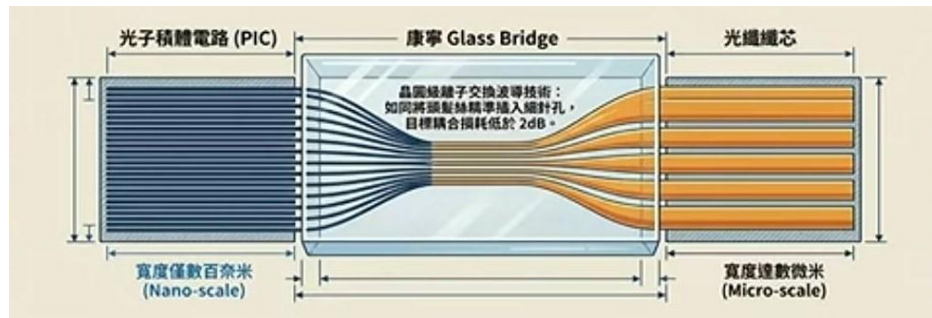
• 支持高密度光子模組：適用於近封裝光學（NPO）、共封裝光學（CPO）和高密度光子模組。

• 這些特點使 GlassBridge™ Fiber-to-PIC Connector 成為未來光學通信系統的理想選擇，能夠有效支持不斷增長的數據需求和技術進步。

Corning GlassBridge: Fiber → Glass Waveguide → PIC Edge Coupler



a) Integrated Module b) GlassBridge™ -PIC interface

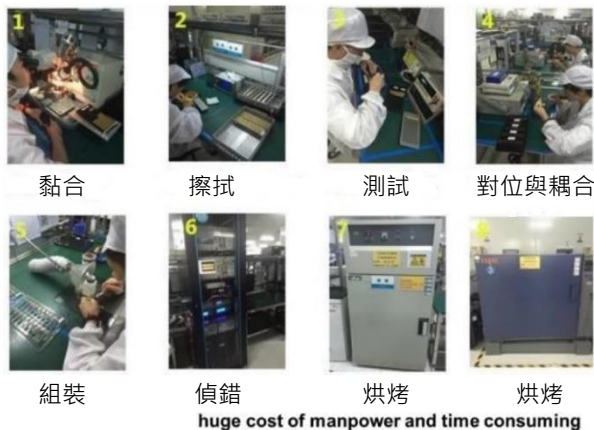


GlassBridge解決光纖與晶片的尺寸與模態匹配問題

技術	GlassBridge 突破	技術價值
Glass Waveguide	離子交換(IOX)玻璃波導	超低損耗、熱穩定性佳
Wafer Bonding	雙層玻璃晶圓鍵合	支援 MT-32 (2×16) 高密度光路
Passive Alignment	利用 Guide Pin + Z-stop 完成被動對位	免主動對位，大幅降低封裝成本
Detachable Interface	採用標準 MT/TMT Ferrule	光纖可插拔、更換容易
Mechanical Isolation	插拔應力與 PIC 耦合面隔離	提升可靠度與壽命

- 解決 Fiber-to-PIC 封裝難題
- 兼具低損耗、高精度與量產能力
- 建立可插拔的標準化光互連介面
- 被動對位技術，推動 CPO 大規模部署

光收發模組生產模式轉移



導入矽光子平台

可自動化項目

- 精準對位
- 對位時間
- 封裝時間
- 組裝良率
- 測試
-

組裝部件數

>40

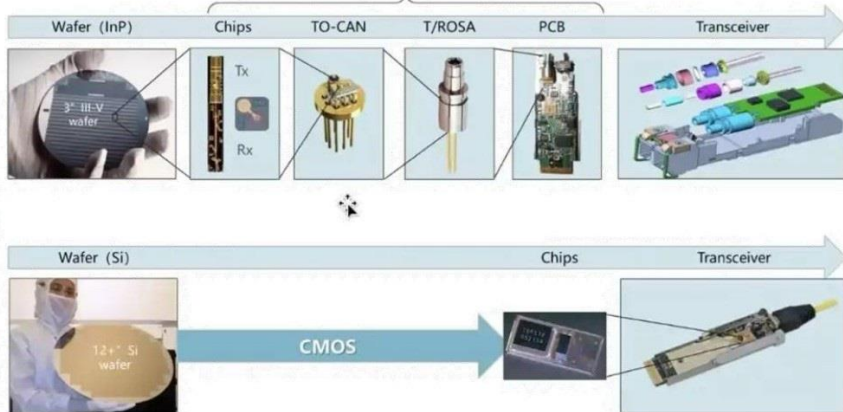
4

PIC, Laser,
Driver IC, Fiber

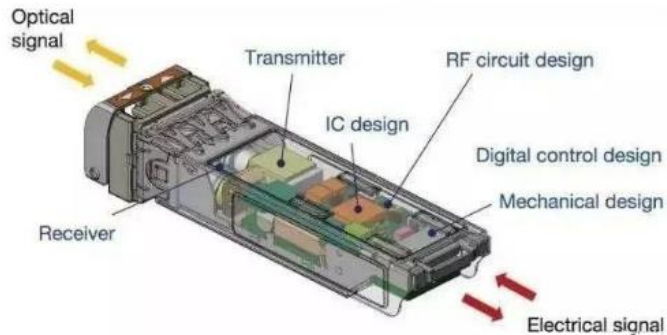
現代矽光子系統要求

封裝層級的要求	策略著眼點
精確的光學對位	將信號損失降至最低
熱穩定性	保持性能的穩定性
高密度整合	支援擴充
製造過程中的可重複性	可實現商業化應用
高效率的組裝方式	促進市場採用

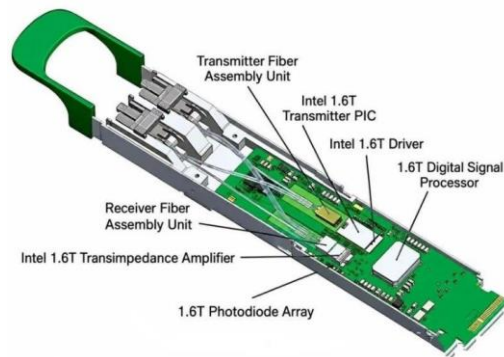
Traditional tech. → Silicon Photonics



成本有望降低20%



800G 單波光收發模組	價值(美元)
DSP*1	90
100G EML*8	88
Driver*2+TIA*2	32
光器件 (TOSA/ROSA)	60
PCB/結構件/殼等其他	55
原材料成本總和	325
人工及加工費	85
總成本	410
總價 (35%毛利率)	631



800G 矽光收發模組	價值(美元)
DSP*1	90
CW(100mW)*2	30
Driver*2+TIA*2	32
矽光芯片	50
PCB/結構件/殼等其他	50
原材料成本總和	252
人工及加工費	75
總成本	327
總價 (35%毛利率)	503

傳統 Pluggable → LPO → XPO → NPO → CPO

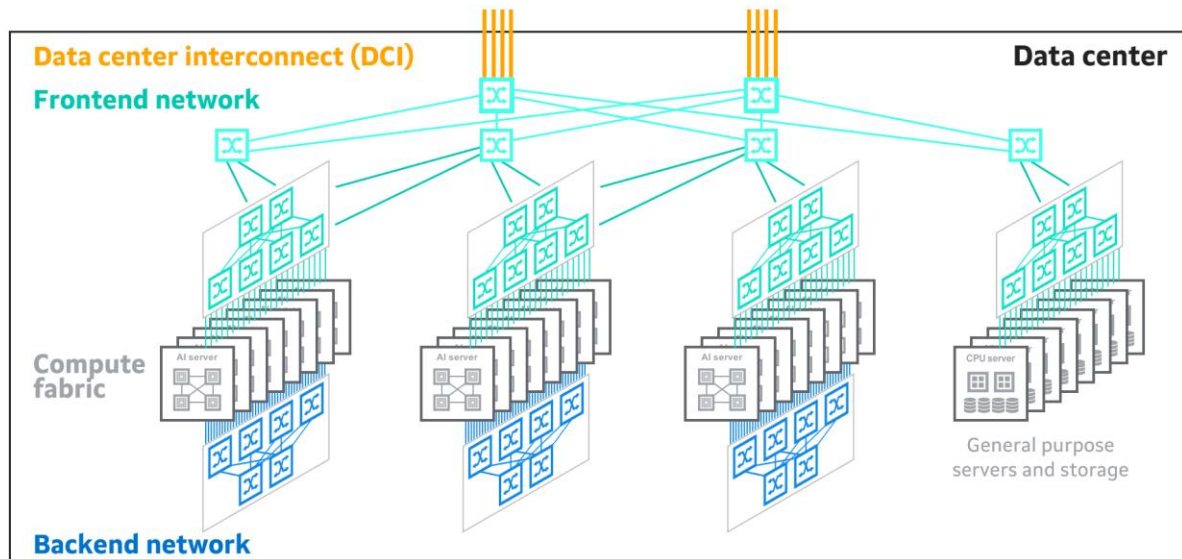
技術項目	全名	DSP 配置	光模組/光引擎位置	電訊號距離	功耗	熱管理	維修性	交換容量	成熟度	傳輸距離	典型應用	架構設計理念
傳統 Pluggable	Pluggable Optics	有完整DSP	前面板	較長	高	容易	最好	受限	成熟	長距離(電信/跨機房)	FTTx、傳統資料中心	光電轉換介面
LPO	Linear-drive Pluggable Optics 線性驅動可插拔光模組	無 DSP 或極簡 DSP	前面板	較長	降低 30~50%	容易	好	中等	快速導入	Rack ↔ Rack	機櫃互連	將DPS功能交給主機
XPO	eXtra-dense Pluggable Optics 超高密度可插拔光學	視方案而定 (傳統DSP或結合LPO技術)	前面板	較長	再降低	中等	中	高	初期、2026最新規範	High-density Switch	超大型資料中心交換機	推升可插拔的物理極限
NPO	Near-Packaged Optics 近封裝光學	極簡DSP	緊鄰ASIC	短	介於LPO與CPO之間	尚可	中上	高	CPO前過渡期	Board ↔ Board/ Rack ↔ Rack	主機板互連、AI高速叢集	讓光引擎緊鄰ASIC，為CPO鋪路
CPO	Co-Packaged Optics 共同封裝光學	幾乎無DSP	與 Switch/ASIC共同封裝	極短	最低	困難	差	最高	逐步商業化	Chip ↔ Chip/ Board ↔ Board	xPU、HBM 高速互連、High-density Switch	將PIC與ASIC/xPU共封裝

這五種架構的演進，是AI 算力中心在面對頻寬需求暴增與功耗/散熱牆限制時，在物理距離、晶片整合度與維護成本之間所做的權衡與拉鋸。

AI需要的4種網路

- Data center interconnect (DCI) network $\approx$ **Scale-across**
- Frontend network connects AI clusters to the cloud data center for storage, switching, and more. CPU has connected to its own optical module. The frontend network always uses Ethernet protocol over optical. $\approx$ **Scale-out**
- Compute fabric connects AI accelerators, GPUs, CPUs, and other components inside servers with interfaces such as NVLink. $\approx$ **Scale-up**
- Backend network uses switches and optical modules to connect the servers into AI clusters.

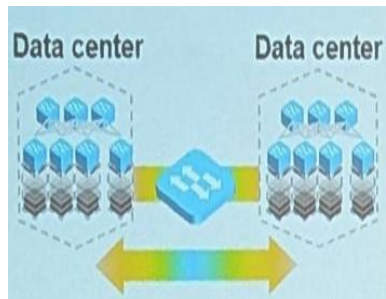
DCI network connects data center to outside world



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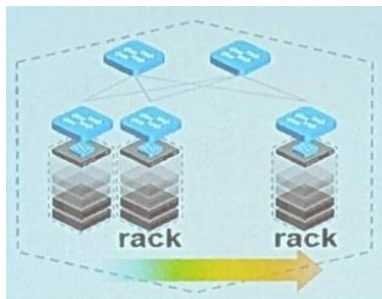
CPO 在不同 Interconnect 場域之應用



Scale-across
Data center Connection

資料中心與資料中心之間的長距離互連；採用 Coherent 與 DWDM communication。

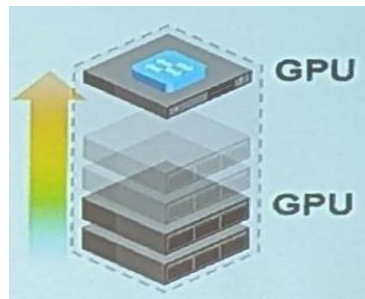
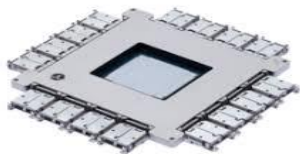
Pluggable Tx/Rx



Scale-Out
Inter-rack Interconnect

利用乙太網路將大量伺服器橫向串聯成大型資料中心網絡。重視CPO與Pluggable相容性CPO。未來也會導入CPO Ethernet Switch。

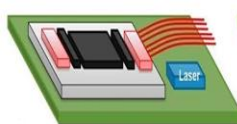
CPO Switch / LPO / XPO



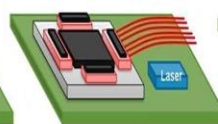
Scale-Up
Intra-rack Interconnect

處理XPU、記憶體之間高密度、多通道互連，如 AI 伺服器內部的XPU (CPU/ GPU/ TPU) 之間器互連。

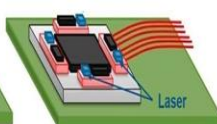
NPO / CPO



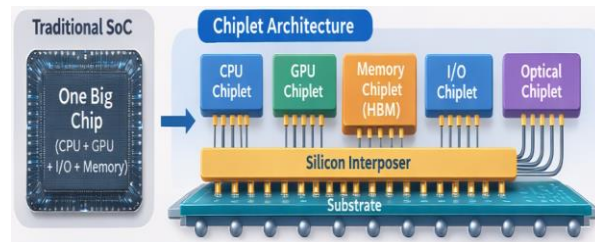
Gen III -2.5D CPO



Gen IV -3D CPO



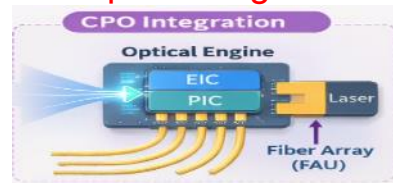
Gen V -Integrated SoC



未來 Scale-In
Inter-chip Interconnect

處理2.5D或3D封裝內晶片與多晶片叢(chiplet, cluster)之間中介層(Interposer)內的傳輸。

CPO / Optical Engine



Infrastructure Core Requirements

Data Center Infrastructure

Inter-Rack : 100Gbps/cable → 400Gbps(主流部署)→ 800G(2025年)→ 1.6Tbps (2026~2027) → 3.2T (2028+)

Inter-Board : 25Gbps/ch → 50Gbps/ch → 100Gbps/ch (2025) → 200Gbps/ch (2027+)

30mW/Gbps → 8mW/Gbps → 3mW/Gbps → 1mW/Gbps (<1 pJ/bit; Future CPO 0.1~0.5pJ/bit)

Inter-Chip : 250Gbps → 1Tbps → 4Tbps → 10Tbps (HBM3 ~819GB; HBM4 ~2+TB/s; GPU-GPU 10+Tps)

2Tbps/cm² → 10Tbps/cm² → 30Tbps/cm² → 100Tbps/cm²

15mW/Gbps → 5mW/Gbps → 3mW/Gbps → 1mW/Gbps

Communication Infrastructure

Metro / Core : 100Gbps/ch WDM → 400Gbps/ch WDM → 1Tbps/ch WDM

Switch : Miniature and Low Power Optical Switch

Access : 10G-EPON → TWDM-PON

MSA協議 Multi-Source Agreement

三大CPO相關的MSA協議

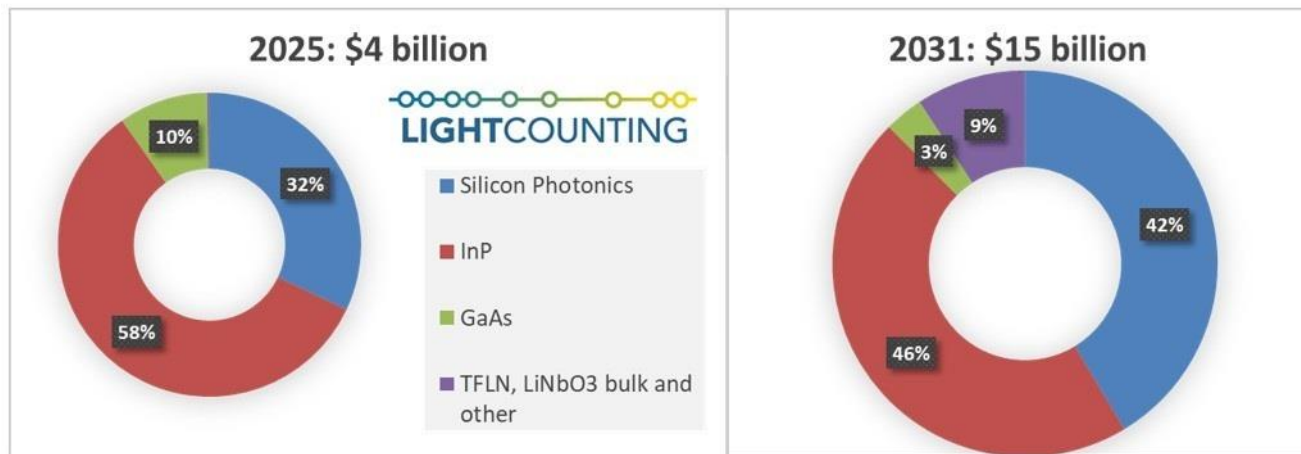
MSA 協議名稱	核心主導與創始成員	針對的關鍵環節與 CPO 定位	技術規格重點
Open CPX MSA		CPO 與內建光學引擎的插座規格。專門定義光學引擎直接安裝在晶片（如 GPU、交換晶片）旁邊的插座（Socket）與機械介面。	定義連接器的機械構造、散熱、電氣腳位（Pinout）等，確保不同廠商的 CPO 光學引擎可以相互汰換。
OCI MSA (Optical Compute Interconnect)	NVIDIA、AMD、Broadcom、Meta、Microsoft、OpenAI	Scale-up（機櫃內部、晶片之間）極短距離光學互連。旨在用光纖取代傳統主機板與機櫃內的銅線。	採用 NRZ 訊號與波長分波多工（WDM）技術。初期為 200Gb/s，未來技術藍圖規劃透過增加波長擴展至 3.2Tbps 以上。
XPO MSA	Arista Networks 主導及各大光模組大廠	Scale-out（機櫃與機櫃之間）外接式大型光模組。處理更長距離、更高頻寬的大型數據傳輸。	支援高達 12.8 Tbps 的極致頻寬，並專門針對液冷資料中心設計可承受 400W 高功耗的整合冷板散熱規格。

- 統一尺寸、外型、電氣、光學、功率、散熱、測試、接口
- 建立生態系、降低壟斷、加速市場普及
- CPO標準化、Optical IO標準化
- MSA 是光通訊產業形成「可互通、多供應商、大規模市場」的核心機制；它不是正式國際標準，但往往比正式標準更快速、更貼近產業落地，是光模組與未來 CPO 生態形成的關鍵。

“光晶片”市場雛形 CAGR 25%

高速光互連所需之光晶片市場

Sales of optical chips used in transceivers, AOCs, LPO/LRO and NPO/CPO by technology



- 2031年，光收發器與光晶片的銷售額將接近 800 億美元。

光晶片：

- AOCs - Active Optical Cable
- LPO - Linear Pluggable Optics
- LRO - Linear Receive Optics
- NPO - Near-Packaged Optics
- CPO - Co-Packaged Optics

Scale-Out → Scale-Up → Scale-In

- 目前：Scale-Out (機櫃、資料中心之間)
- 未來：Scale-Up (機櫃內、GPU/GPU 高頻寬互連)
- 再下一階段：Scale-In (封裝內 Chiplet 光互連)

- InP CW Laser、可調雷射、InP PIC需求成長
- 矽光子 + InP 異質整合成技術主流
- 薄膜氮化鋁、GaAs也同步受惠

FAU & Coupling 挑戰與解決方案

Theory, Technical, Engineer Challenges

Fundamental Technologies

Photonics and Electronics Integration Technologies

■ Integration Technology

- **Photonics and Electronics Monolithic Integration**
- **Light Source / Semiconductor Optical Amplifier (SOA) Integration**
- High Temperature Operation
- WDM, Multichannel Integration

■ Device Technology

- Nano Photonics
- **Ultra Miniature Modulator**
- **Nano Laser**
- Stacking, 3D Structure

Manufacturing Technologies

- Packaging
- Process, Fabrication
- Design Tool, Evaluation, Automation and SiPh Platform

FAU光耦合公差精度挑戰

光耦合技術的「產業四大門檻」

1 Coupling Efficiency $\geq 70\%$

- 低效率將直接影響：系統功耗、訊號品質
- 設計能力（光學 + mode matching）是核心

2 Yield（良率） $\geq 90\%$

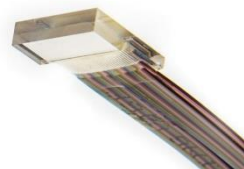
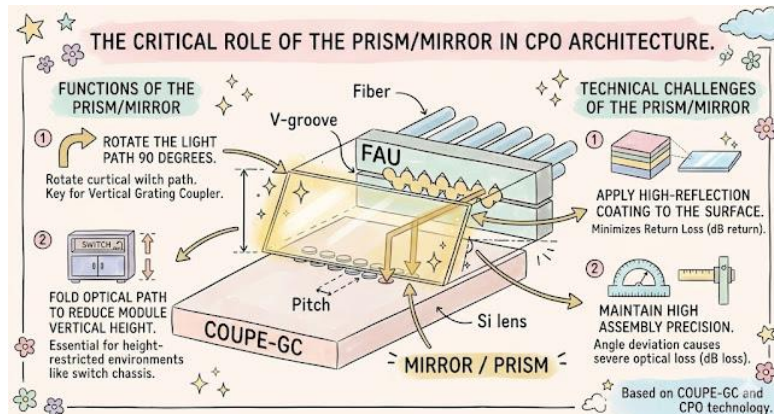
- 必須有量產技術、不能依賴精密半自動對位

3 組裝 Cycle Time 1~2 分鐘

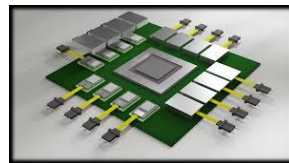
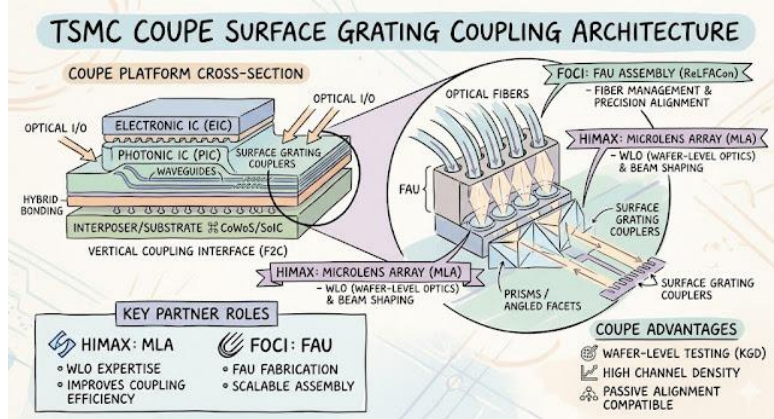
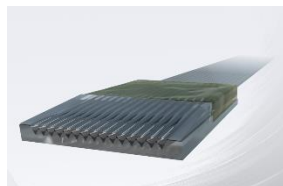
- 每顆元件組裝時間需極短
- 適用於：自動化產線、高產能需求

4 Insertion Loss ≤ 0.5 dB

- 系統總損耗需極低
- 對高速光通訊特別關鍵



V-Groove Fiber Arrays



大立光電 FAU解決方案



LARGAN Precision

#1 high end lens supplier for mobile phone.

Industry leader on automated manufacturing with in-house automation R&D team.

CPO Solutions

PMLA Multi-row

Fiber Array Multi-row

Contact Info:
Ray Han, Account Manager rayhan@largan.com.tw
Jones Yen, Account Manager jonesyen@largan.com.tw

大立光電 LARGAN

QR code

1. 左圖為大立光的FAU、Prism Micro Lens Array (PMLA) 的產品。兩者搭配，將光耦合進PIC裡面。
2. 大立光表示所有的設備都是自行開發，包括高密度FAU、多層V-Groove結構、微米級對位、自動化封裝。
3. 大立光表示，有在進行 Metalens 的研究。



Metallens 解決 Coupling 問題

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Meta-lens for co-package optics and fiber array unit coupling

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Jenq-Yang Chang and Mount-Learn Wu
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Author e-mail address: *flicar@authenx.com

Abstract: We demonstrate a meta-lens assisted co-package optics which enables multi-channel detachable fiber array unit to silicon photonics chip coupling and results in 1dB alignment tolerance up to $\pm 18\mu\text{m}$. © 2025 The Author(s)

1. Introduction

As the demand for high-bandwidth and energy-efficient data transmission continues to grow, co-packaged optics (CPO) has emerged as a promising solution for next-generation optical interconnects [1]. CPO integrates photonic and electronic components within a single package, significantly reducing power consumption while improving signal integrity. However, achieving reliable and scalable optical coupling between photonic integrated circuit (PIC) chips and fiber array units (FAUs) remains a major challenge, particularly for detachable interfaces and high-volume manufacturing [2].

This study presents a novel meta-lens-assisted coupling architecture designed to overcome these limitations. By employing a pair of meta-lenses, one mounted on the PIC and the other on the FAU, the system enables efficient beam collimation and focusing, thereby facilitating detachable optical connections with enhanced alignment tolerance [3, 4]. The meta-lens structures are carefully optimized to achieve superior collimation quality. This approach not only enhances coupling performance but also demonstrates strong potential for scalable manufacturing, making it well-suited for future optical communication systems.

2. Design of meta lens for CPO applications

2.1 System Design Overview

Two types of optical couplers are commonly used to facilitate signal transmission between silicon photonic chips and optical fibers: grating couplers (GCs) and edge couplers (ECs). This work specifically focuses on optical coupling between GCs and single-mode fibers (SMFs). To achieve detachable functionality, the optical system is designed to generate a collimated beam with a diameter of approximately $70\mu\text{m}$, which significantly improves the alignment tolerance to $\pm 18\mu\text{m}$. The system consists of two meta-lens assemblies: the lower assembly integrates a meta-lens mounted on the PIC, while the upper assembly incorporates a meta-lens attached to the FAU, as illustrated in Fig. 1(a). The PIC acts as the carrier substrate and was fabricated through a standard silicon photonics foundry process. It supports multi-channel I/O using conventional O-band grating couplers. The 2D FAU used in this work comprises 12 rows and 3 columns, totaling 36 channels, with pitches of $500\mu\text{m}$ (horizontal) and $250\mu\text{m}$ (vertical). The FAU integrates both SMFs and polarization-maintaining fibers (PMFs) to enable signal transmission and external light source (ELS) input. The first meta-lens collimates the light emitted from the GC vertically, while the second meta-lens focuses the collimated beam into the SMF. Both meta-lenses are fixed in place using epoxy adhesive, and a mechanical fixture is employed to maintain positional stability during the mating process of the two assemblies. Figure 1 schematically illustrates the concept of meta-lens-assisted detachable coupling between the PIC and FAU.

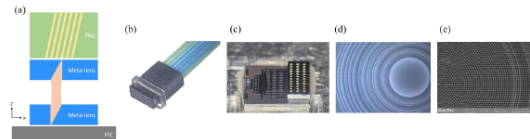
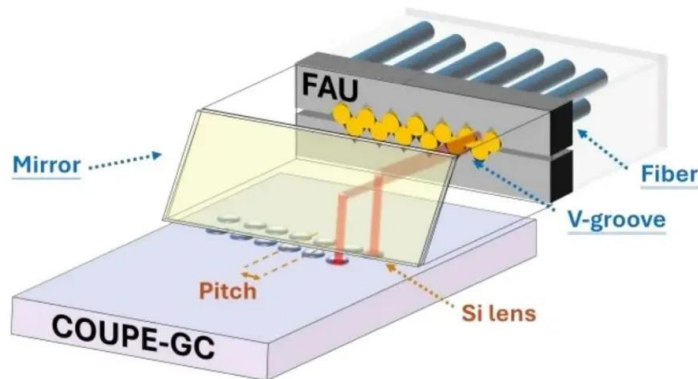


Fig. 1. (a) Schematic of the meta-lens-assisted CPO-to-fiber coupling architecture. Meta-lens mounted on (b) the FAU end face and (c) the PIC surface. (d) Optical microscope image and (e) SEM image of the fabricated meta-lens.

2.2 Design Methodology



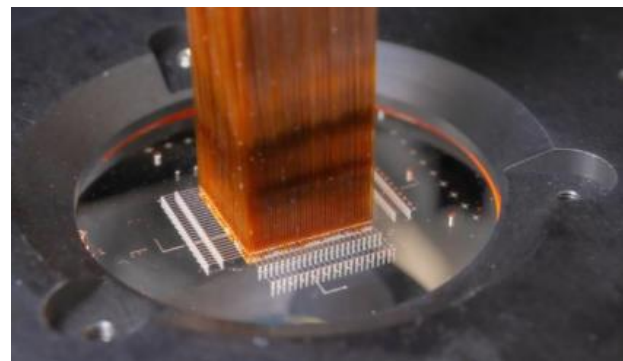
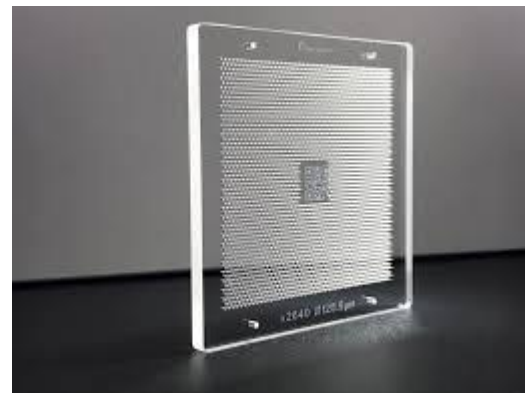
在晶片與光纖端各放一片 Meta-lens，將原本發散的光束轉化為直徑約 $70\mu\text{m}$ 的準直光，而大幅放寬對準要求。

- 耦合機制：系統由兩組 Meta-lens 組成。(a)底部的 Meta-lens 封裝在光子積體電路 (PIC) 上，負責將光柵耦合器 (GC) 射出的光準直化；上方的 Meta-lens 則貼合在 FAU 上，負責將準直光聚焦進入單模光纖 (SMF)。
- AuthenX 這項技術之所以值得關注，是因為它解決了 CPO 從實驗室走向工廠的三大路障：量產性、維護性與穩定性。

瑞士 Femtoprint

Fiber connectivity goes 2D and sub-micron alignment precision

- 雷射微加工玻璃 ferrule，做 through-hole array
- Femtoprint 可利用 fused silica (石英)、Borofloat (硼氟玻璃-BF33)、超低膨脹玻璃/ (Ultra-low Expansion-ULE glass)，以 femtosecond laser micromachining 加工 10×10 垂直孔，再把 Corning SMF-28 類標準光纖逐根穿入、主動對位、UV 膠固定。
- Femtoprint 宣稱可做 2D hole array glass ferrule，孔參數可控制到約 $0.5 \mu\text{m}$
- 量產批次標準差低於 $0.5 \mu\text{m}$ ，且可做 fiducial / alignment mark。

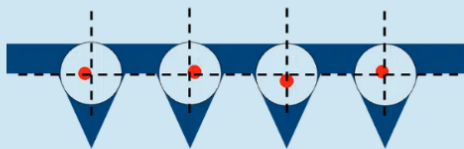


荷蘭 MicroAlign

High-accuracy fiber arrays with core-active alignment

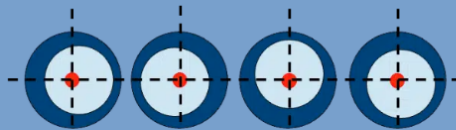
- 微孔玻璃/陶瓷板 + 主動式 core alignment
- MicroAlign 宣稱可用 MEMS 工具逐根調整 fiber tip，達到 core positioning <100 nm RMS (均方根)。

在基於 V 型槽芯片技術、並採用被動式對準方式的纖維陣列中，光纖陣列與光子晶片之間的最佳耦合效率，會受到纖維陣列中核心位置的精確度的限制。



V-groove (passive alignment)

MicroAlign 所生產的光纖陣列中，各個光纖的核心都經過了精確的調整，從而確保了最高的精度。這種設計使得光纖陣列與芯片之間的連接更加高效。

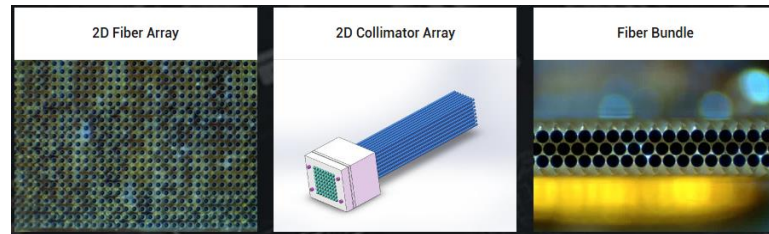


MicroAlign (active alignment)

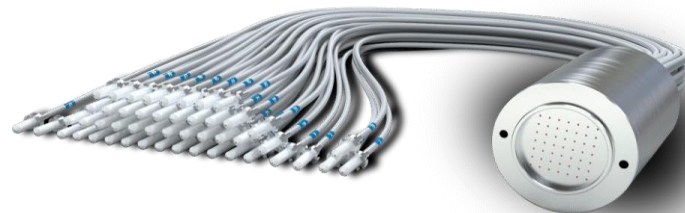


Other FAU

- 中國中山市美素光學科技 (MEISU Optics)：精於光纖陣列 (Fiber Array) 與光學設備的研發與製造；包括 2D Fiber Array 170、Fiber Collimator 和 Fiber Optic Bundle。
 - Pitch Tolerance for 2D Fiber Array: $\pm 1\mu\text{m}$ or $\pm 2\mu\text{m}$ depend on glass faceplate or ceramic one.



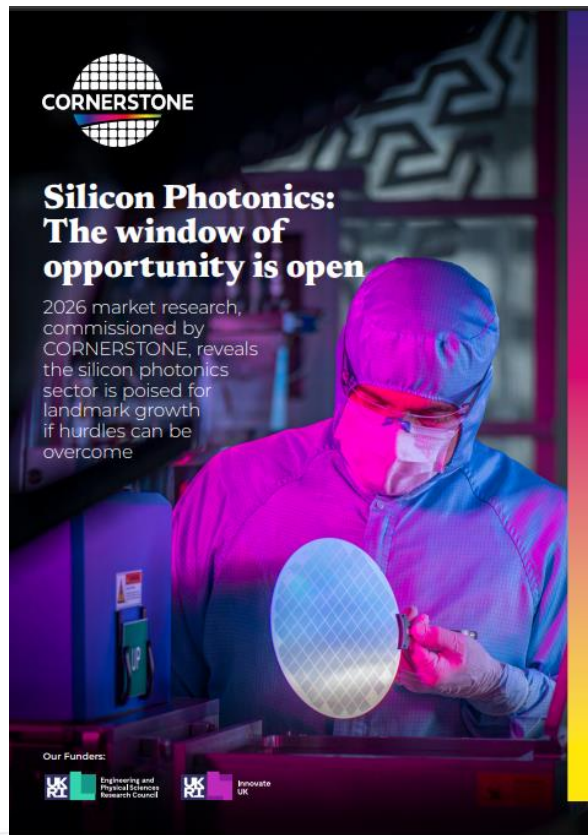
- 法國 IDIL Fibres Optiques：提供多種類型的光纖陣列，包括單模、多模、保偏光纖等；精度可達 ± 1 毫米、光纖核心之間間距精度可控制在 1 微米以內。
 - Pitch accuracy $\pm 1\mu\text{m}$
 - Fiber length accuracy $\pm 1\text{mm}$
 - Micro-polishing accuracy $\pm 0.5^\circ$



挑戰：矽光子 & InP之異質整合

- Challenges：Coupling external light sources into the PIC、lighting leakage、Indium Phosphide (InP) Integration、Getting light out of the PIC、Optimizing the performance of the PIC itself、EDA premature
 - InP（磷化銦Indium phosphide，InP）和矽與電光半導體等其他材料相結合的混合光子積體電路(PIC)是關鍵。
 - 材料、設計和製造方面仍然面臨著重大挑戰。
 - 要實現這些多樣化的應用，重要的是與不同材料（半導體和其他）與矽做整合。
 - 但目前最大的挑戰是發展新的製程工藝，但也得適應可能少量多樣的矽光子應用市場。
- 調變器還是得運用鈮酸鋰、鈦酸鋇等其他材料，故有幾家公司正在開發鈮酸鋰材料平台。
- 新創公司正在尋求第三個目標：將光源整合在晶片上。
 - 英特爾將磷化銦多層堆疊，以便將雷射、SOA整合到同一平台上。”
 - 另一種新方法是微轉印(Micro-transfer printing)，可將雷射、光放大器轉印到矽平台。
- 會有兩個不同的參與者：矽晶圓代工廠、III-V代工晶圓。誰將成為負責整合的產業主導者？也許是在矽晶圓廠？

Cornerstone report: 矽光子產業開發情況



Southampton, UK, 17 June 2026 – Cornerstone 最新市場研究顯示，量產的挑戰若不盡快解決，會阻礙矽光子 (SiPh) 的成長。這項調查的對象為英、美、荷、德的500位決策者；他們正在開發SiPh相關技術。研究指出，目前阻礙該產業發展的諸多障礙。研究也強調，如果能夠克服這些挑戰，將會創造大量的就業和收入機會。

- 66%的受訪者認為量產是商業化的主要障礙。
- 59%的受訪者表示本國缺乏足夠的產業生態系來支持從研究到量產的過程。
- 67%的受訪者認為量產能力對於增強國家技術自主性至關重要。
- 27%表示，代工廠漫長的交貨週期導致了延誤，而關稅等地緣政治因素則進一步加劇了全球供應鏈的壓力。
- 31%表示他們延宕了產品開發計劃，平均損失估計為270萬美元。
- 54%認為現有的政策對矽光子的支持力度不足。
- 42%的英國企業表示面臨人才短缺；24%的企業表示，在過去兩年裡，他們的員工流失到了海外市場。

幾種光電整合(PIC)平台之比較

	InP	SiPh	SiN	Glass	Polymer	Silica	LiNbO3
Passive components	++	++	+++	+++	+++	+++	Hybrid
Polarization components	++	++	++	+	+	Hybrid	Hybrid
Lasers	+++	Hybrid	Hybrid	Hybrid	Hybrid	Hybrid	Hybrid
Modulators	+++	++	+	Thermal	+++	Hybrid	++++
Switches	++	++	+	+	+	+	Hybrid
Optical amplifiers	+++	Hybrid	Hybrid	Hybrid	Hybrid	Hybrid	Hybrid
Detectors	+++	++	Hybrid	Hybrid	Hybrid	Hybrid	Hybrid
PROs	Best for laser integration	Best for electronic/optical integration	Low cost Small size	Simple process, low cost	Compatible with Si/InP platforms	Low losses Low cost	Very good modulation function
CONs	High cost and low yield for components integrating other elements.	No light generation	Material properties are process-dependent	Few functions are possible	Reliability / thermal management issues	No active functionalities	Not a thin-film tech yet
INDUSTRY STATUS	HIGH-VOLUME LASER	RAMPING UP TO high-volume transceiver	LOW-VOLUME PRODUCTION	PRE-SERIES	R&D/ QUALIFICATION	HIGH-VOLUME couplers	HIGH-VOLUME modulators

玻璃基板 (Glass Substrate) 崛起

Glass Substrates and Silicon Photonics: Potential Building Blocks for Future Optical I/O Systems

Intel's Rio Rancho Initiative – Illustrative Future Integration Concept (Not a Disclosed Product)

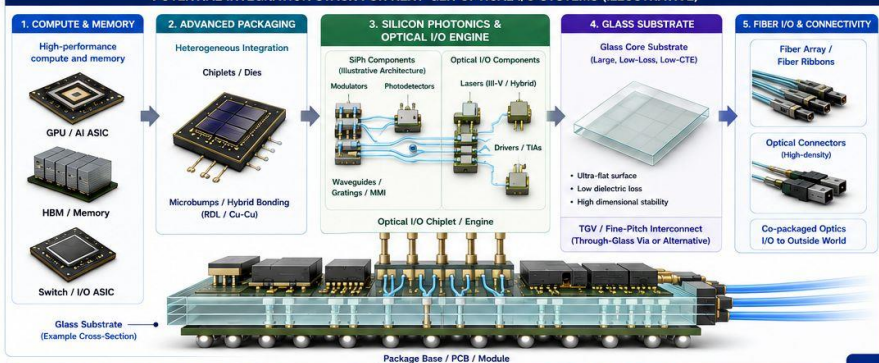


RIO RANCHO
NEW MEXICO, USA



- Intel is reportedly targeting the world's first glass substrate output at Rio Rancho.
- Intel is offering silicon photonics manufacturing services to external foundry customers.
- Glass substrate + CPO prototypes have been demonstrated.
- Industry estimates suggest commercialization toward 2030.*

POTENTIAL INTEGRATION STACK FOR NEXT-GEN OPTICAL I/O SYSTEMS (ILLUSTRATIVE)



Why Glass Substrates?

- Better Signal Integrity**
Lower loss and smoother transmission for high-speed electrical & optical signals
- Potential Thermal & Dimensional Advantages**
Lower CTE, improved dimensional stability and package flatness
- Higher Interconnect Density**
Finer lines/spaces and more routing options for advanced packaging
- Excellent Flatness & Stability**
Enables larger package sizes with improved yield and reliability
- Sustainability Potential**
Recyclable and more environmentally friendly material

RIO RANCHO ROADMAP (RUMORED)

- 2024-2025
• Si Photonics Manufacturing Services
Offering SiPh PDK and manufacturing to external foundry customers
- 2025-2026
• CPO / Optical Engine Prototypes
Glass substrate + CPO demos already shown
- 2027-2029
• Glass Substrate Pilot Production
Ramp and yield learning at Rio Rancho
- ~2030
• Industry Estimates:
Commercialization Toward 2030*

* Based on industry reports and analysis.

KEY BENEFITS FOR AI INFRASTRUCTURE

- Higher Bandwidth**
Enables >1.6T and beyond per channel
- Lower Power**
Reduced electrical I/O power and better energy efficiency
- Better Thermal Management**
Supports higher power density AI packages
- Higher Reliability & Yield**
Lower warpage and improved long-term stability
- Sustainable Optical I/O**
CPO + SiPh + Glass enables scale-up and scale-out

POTENTIAL USE CASES



ECOSYSTEM COLLABORATION (ILLUSTRATIVE EXAMPLES)

- Materials & Substrate (Glass, TGV, Interposer)
 - Photonic Devices (Lasers, Modulators, Detectors)
 - Assembly & Packaging
 - Fiber & Connector Ecosystem
- Working with customers and ecosystem partners to build a complete optical I/O solution



The next wave of AI infrastructure will be built on the convergence of advanced packaging, silicon photonics, and innovative substrates—enabling higher performance, lower power, and greater scalability.

Note: This is an illustrative concept based on public information and industry analysis. Not a disclosed Intel product or roadmap.

Source: TrendForce, May 26, 2026
(Information based on reports and industry analysis; subject to change.)

未來的AI基礎設施依賴先進封裝、矽光子，以及創新基板的整合。玻璃基板可建構未來光學I/O 與延續CPO規模化發展之基礎：

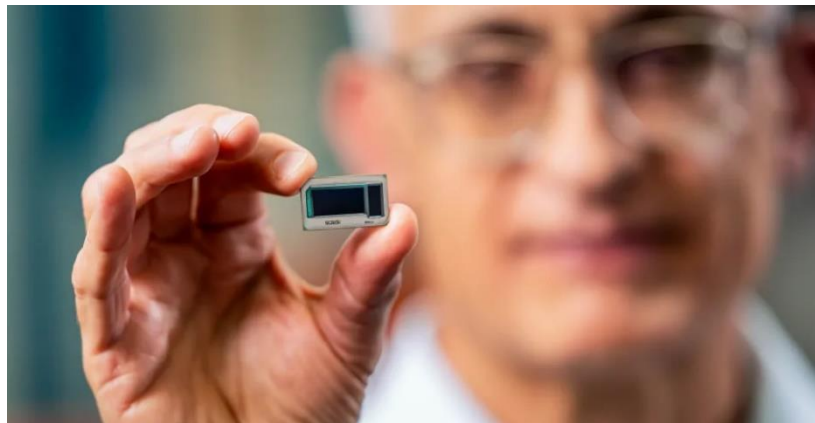
1. 卓越的平整度與穩定性
2. 熱與尺寸優勢
3. 滿足矽光子高精度對準的需求
4. 支援更細的線寬線距與更多佈線選擇，提供更高的光互連密度。

可能的時程：

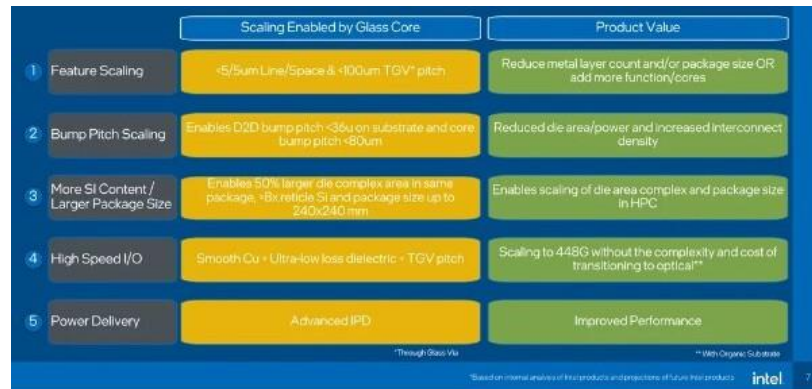
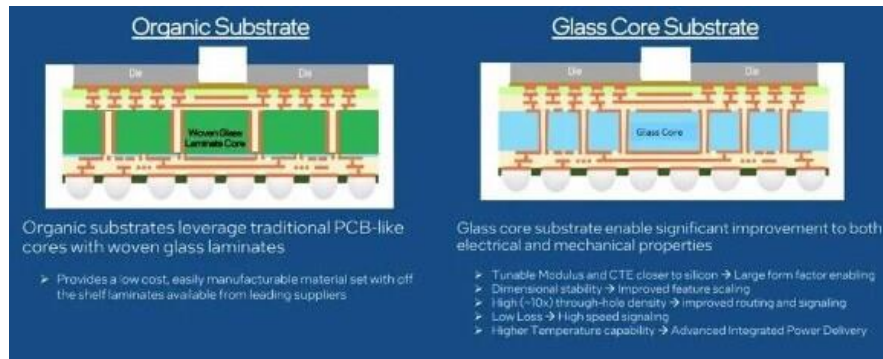
1. 2025~2026年：產業已展示「玻璃基板 + CPO」的雛型晶片 (Prototypes/Demos) 。
2. 2027~2029年：玻璃基板的試產與良率提升。
3. 2030年~：邁入全面商業化量產。

Summary：傳統基板已達極限。玻璃基板可成為將GPU/ASIC、HBM，以及矽光子晶片進行異質整合之基板。

Intel's Glass Substrate



- 互連線寬/線距： $< 5/5 \mu\text{m}$ 、玻璃穿孔間距： $< 100 \mu\text{m}$
- 基板上晶片對晶片凸塊間距： $< 36 \mu\text{m}$; Core bump pitch: $< 80 \mu\text{m}$
- 玻璃基板可以比現有的PCB減少超過1/4厚度
- 極高平整度：改善曝光製程中的焦點景深
- 比傳統基板減少 50% 的圖案形變
- 極佳的幾何維度穩定性與層間對準精度
- 耐溫特性：可電感、電容等元件直接嵌入玻璃中，整合光互連。



積體光電之應用領域以及合適之平台

Applications		Telecom	Datacom	Wireless communication	Automotive interconnects	Sensors	Medical
Product examples		Coherent optical transceivers AWG Modulators	Optical transceivers (100G/400G) Embedded optics (200G)	Optical transceivers (25G)	Optical transceivers for intra-car interconnects	LiDAR Gas sensors	OCT Immunoassay
Typical wavelength		1310 – 1550 nm	1310 – 1550 nm	1310 – 1550 nm	n.a.	900 – 7000+nm	1000 – 1500 nm
PIC platform	SiPh	X	X	X	X	X	X
	InP	X	X			X	
	SiN					X	X
	Polymer	X					
	Glass		X			X	X
	Silica	X	X				
	LiNbO3	X					

對於光的理解：From Geometrical to Physical Optics

巨觀

Geometrical Optics

- Light: Rays
- Intensity (cd, lm)

Optics



Photonics

How to Bridge??

微觀

Physical Optics

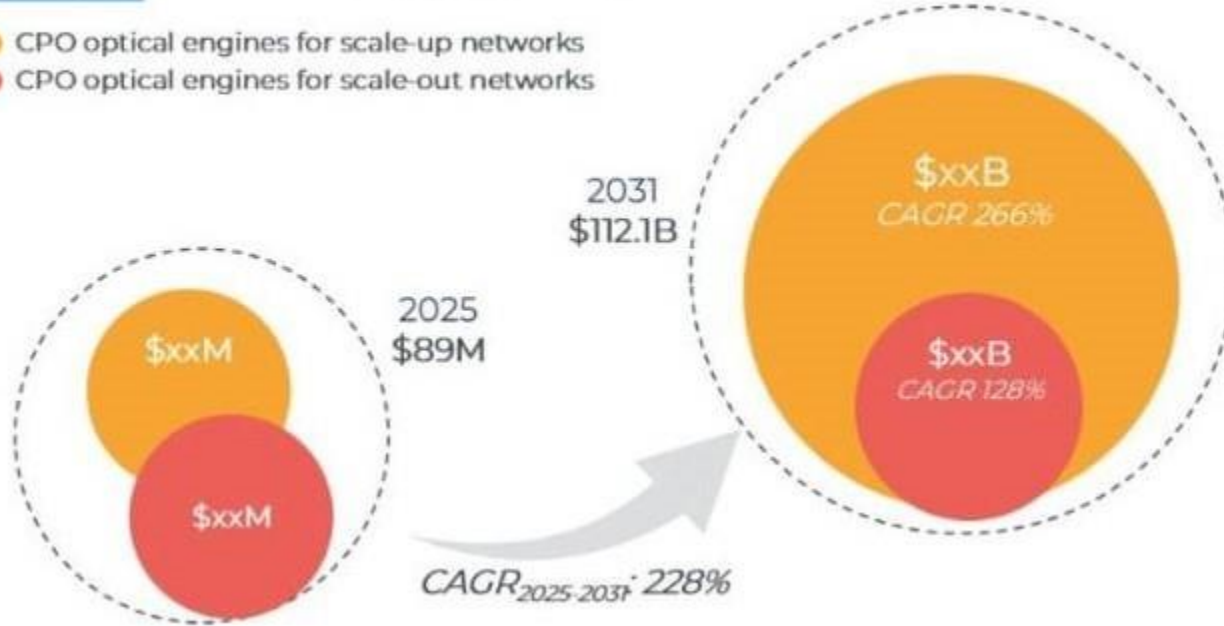
- Light: EM waves, Maxwell Eqs.
- Energy: Poynting Vector $\mathbf{S} = \mathbf{E} \times \mathbf{H}$
- Phase
- Polarization, TE, TM
- Degree of coherence)
- Interference, diffraction

CPO Market

2025-2031 CPO optical engines revenue growth forecast by application

Co-packaged Optics for Data centers 2026, July 2026, Yole Group

- CPO optical engines for scale-up networks
- CPO optical engines for scale-out networks



Company	Silicon Photonics Revenue (USD Bn)	Implied Market Share
Broadcom	\$1.14 Bn	17.6%
Coherent Corp.	\$0.88 Bn	13.5%
Cisco Systems (Acacia)	\$0.68 Bn	10.5%
Intel	\$0.64 Bn	9.9%
Marvell Technology	\$0.56 Bn	8.6%
InnoLight Technology	\$0.40 Bn	6.2%
Lumentum Holdings	\$0.38 Bn	5.9%
Molex	\$0.30 Bn	4.6%
Ayar Labs	\$0.12 Bn	1.9%
GlobalFoundries (Photonics Products)	\$0.10 Bn	1.5%
STMicroelectronics	\$0.07 Bn	1.1%
Hamamatsu Photonics	\$0.07 Bn	1.0%
Ranovus	\$0.06 Bn	0.8%
DustPhotonics	\$0.05 Bn	0.7%
Other regional manufacturers	\$1.05 Bn	16.2%

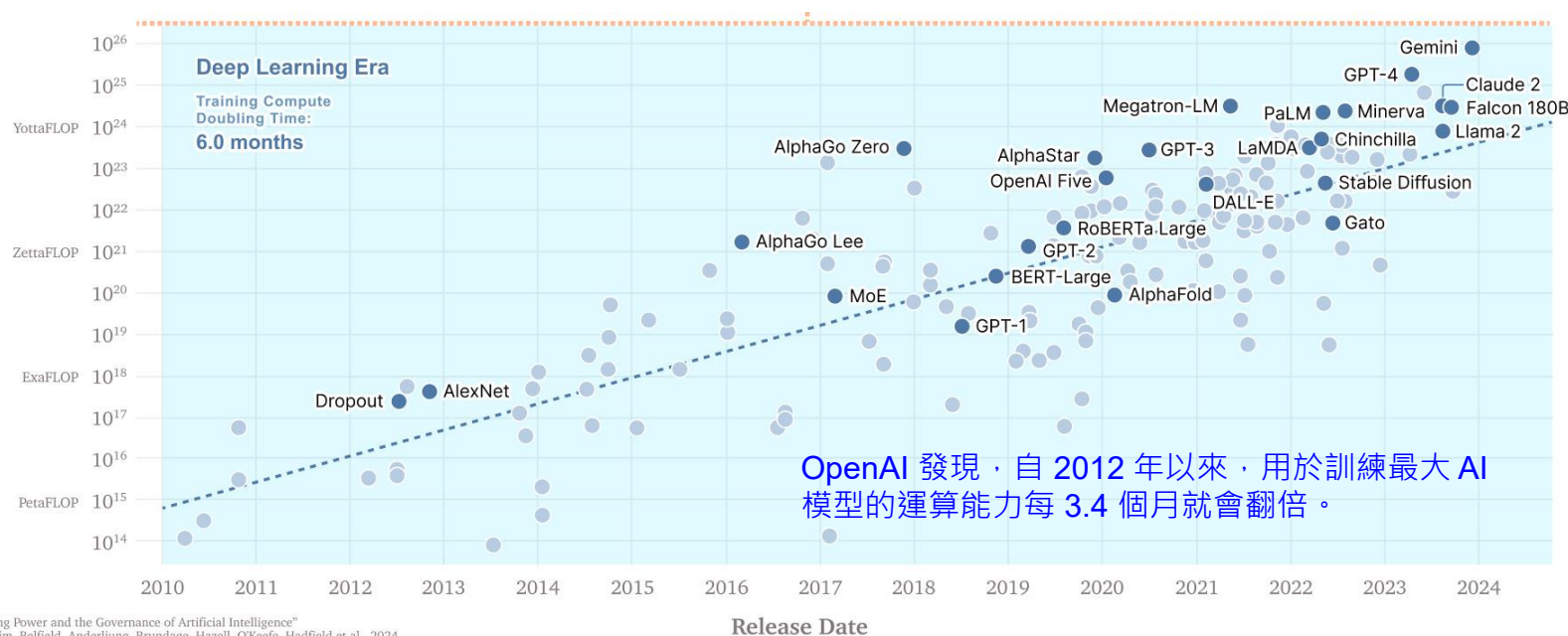
- Beyond Communication -

Silicon Photonics for Computing Application

Optical Computing for AI accelerator

整個資通訊產業都在支援AI算力的需求

The Exponential Growth of Floating-point Operation for AI Need



AI算力與互連需求擴大整體ITC產業

若AI算力需求增加為10倍，對ICT週與產業鏈之放大效果	營收放大
AI GPU / ASIC / HBM / CPO高速互連	5 ~ 10倍
AI伺服器、交換器、電源、液冷、機櫃	3 ~ 6倍
資料中心CapEx / AI資料中心建設	2 ~ 5倍
半導體整體市場	1.5 ~ 2.5倍
整體ICT產業：PC、手機、企業IT、電信、軟體、消費電子等	1.2 ~ 2倍

- IDC估計2026年全球半導體市場可達1.29兆美元，年增52.8%，主要由AI基礎建設、記憶體推動。
- 如果AI算力10倍 → AI基礎建設3 ~ 6倍 → 半導體1.5 ~ 2.5倍 → 整體ICT約1.2 ~ 2倍。
- GPU、HBM、先進封裝、光通訊、電源、冷卻、資料中心網路這些「瓶頸環節」會被高度放大。
- 如果AI算力10倍，局部領域，例如HBM、CoWoS/先進封裝、800G/1.6T光通訊、CPO、液冷、AI電源模組，成長幅度可能接近或超過10倍。
- 讓ICT產業結構重新分配，價值集中到算力、記憶體、互連與能源基礎設施。

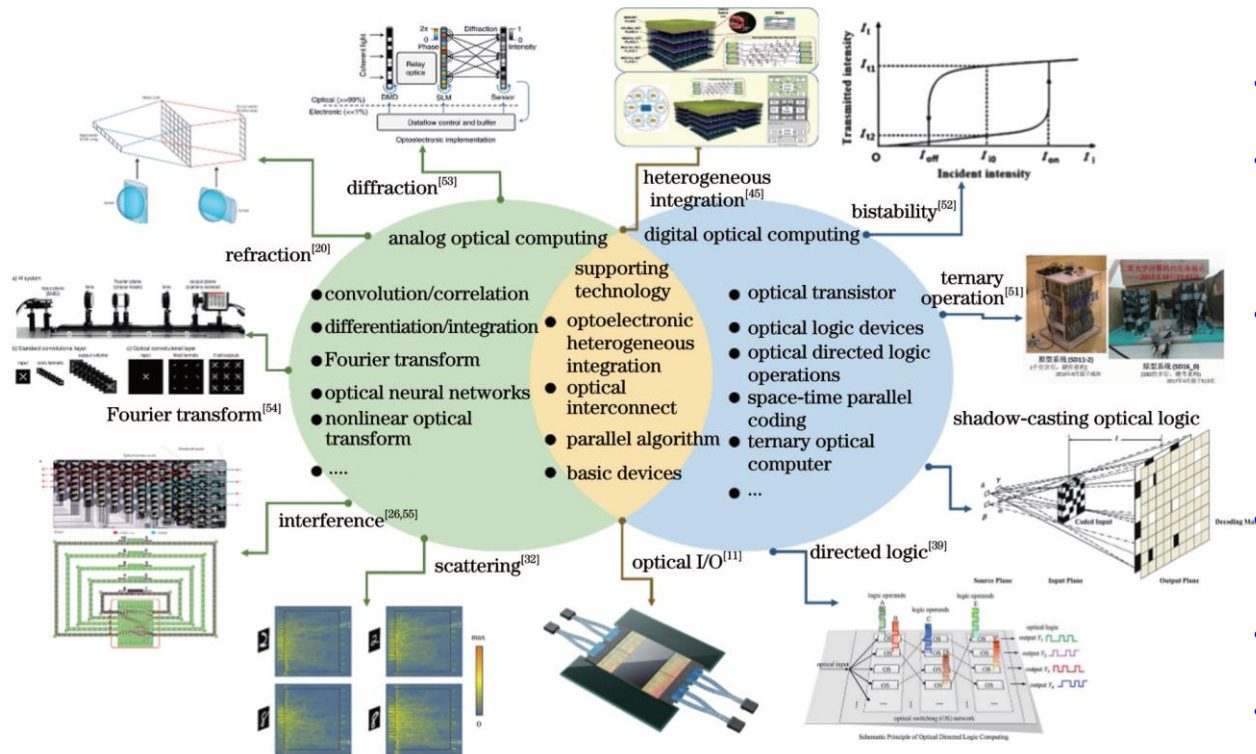
Analog vs. Digital Optical Computing

類比優點

- 資訊密度高，無量化誤差。
- 光電材料與微納結構設計自由度高。
- 運算速度快。
- 與深度學習演算法匹配。（AI接受近似解）

類比缺點

- 訊號受級聯系統雜訊影響較大，需要開發新的容錯方法。
- 特定的資訊處理。
- 高運算精度要求系統具有高線性度和動態範圍。



數位優點

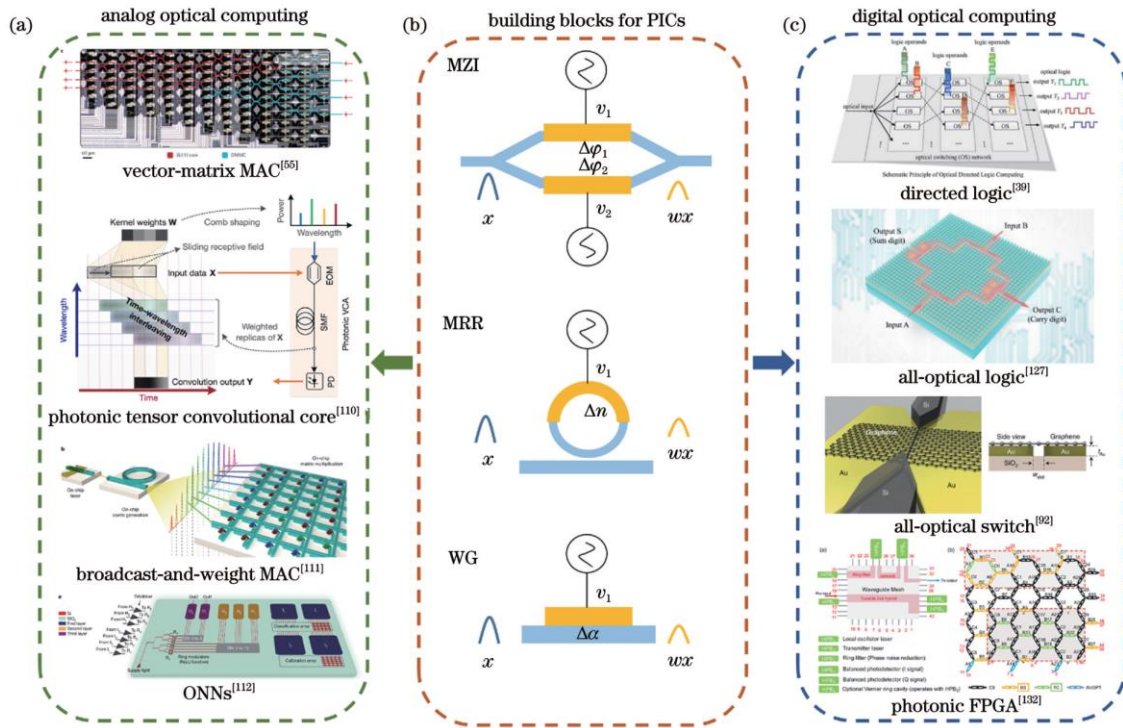
- 無級聯噪音，抗干擾能力強。
- 計算結果可以方便地移植到現有的電子數位設備中進行儲存和處理。
- 在同一系統下，透過編碼可以獲得更高的計算精度。

數位缺點

- 時間和空間解析度低，計算規模受到限制。
- 訊號處理演算法複雜度高。
- 每個編碼位元都必須精確計算，對系SNR要求較高。

矽光子可做為平面積體光學運算之製作技術方案

平面整合式光學計算基本單元及相關技術方案



平面積體光學計算主要單元：

- 馬赫-曾德爾干涉儀陣列 (MZIs)
- 微環諧振器陣列 (MRRs)
- 波導調製器陣列 (WMA)

實現向量-矩陣乘法 (VVM-MAC) 或

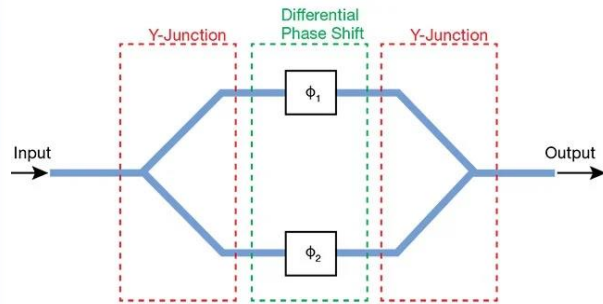
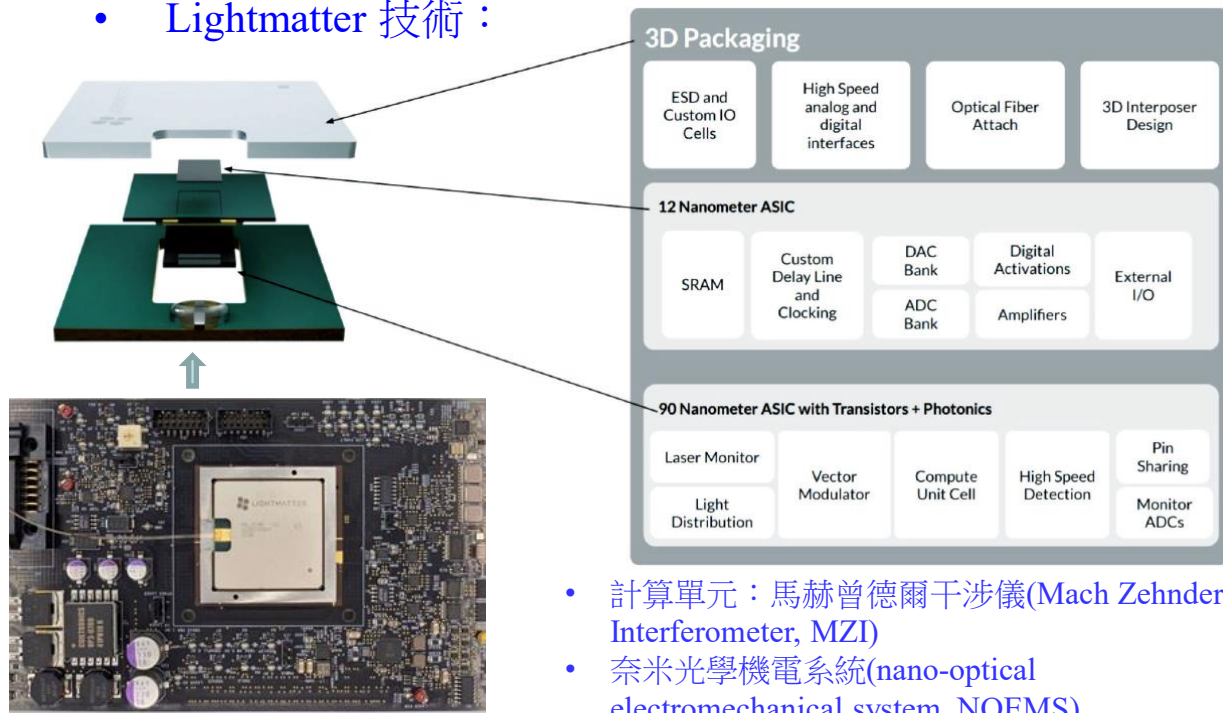
ONNs

平面式類比光學運算之分類

- 同調式
- 非同調式

Benchmark: Si-Ph 光學計算/運算-推論/卷積新創公司紛起

- Lightmatter, Lightelligence, Ayar Labs, Intel, Global Foundry, Akhetonics
- Lightmatter 技術：



Lightmatter:

- 其晶片能效是Nvidia A100的20倍，處理量則至少是5倍。
- 光子處理器具有一個 64×64 光子矩陣向量乘積運算器(photonic matrix vector product calculator, vector-matrix multiplication, VMM)；資料可以在200 picoseconds於晶片內傳播。

- 2025 Lightmatter expands photonic AI interconnect deployments; cumulative funding exceeds \$850M, with valuation reaching approximately \$4.4B.

Benchmark: 曦智科技Lightelligence推出AI加速器



A股港股 港股最狂IPO！AI算力熱潮引爆 曦智科技首日暴漲400% 沈亦晨十年打造光計算獨角獸

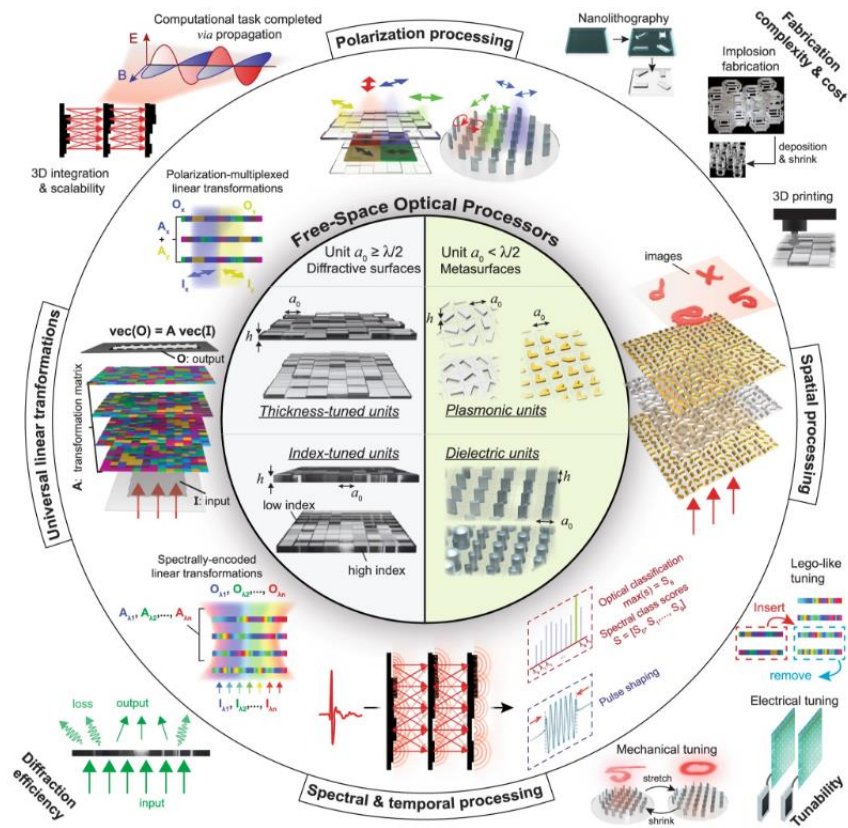
鉅亨網新聞中心 | 2026-04-29 07:10

港股市場再現資本狂熱。28日上午9點20分，香港交易所兩面銅鑼同時敲響，分別迎來二次上市的邁威生物(02493-HK)與被稱為「AI光算力第一股」的曦智科技(01879-HK)。不過兩者首日表現明顯分化，邁威生物漲幅不足1%，而曦智科技股價暴漲逾400%，收報905港元，對比發行價183.2港元，市值飆升至約832億港元，成為今年港股IPO首日漲幅最高個股。



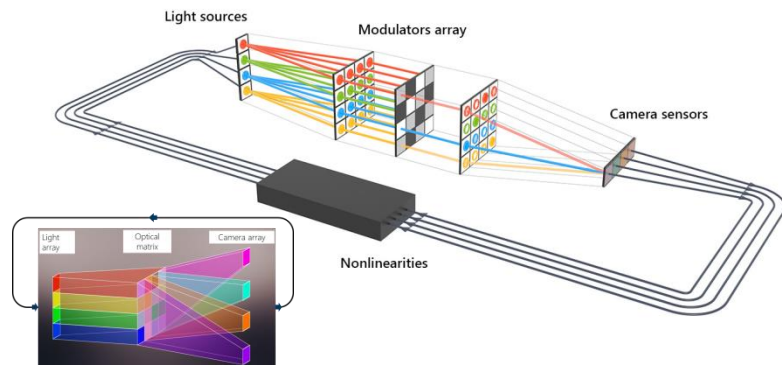
曦智科技執行長沈亦晨指出，光子晶片代表一個更快實現產業化的技術方向，因為光子晶片不依賴製程、技術也不受限，可用 28 奈米的電子晶片產生 7 奈米電子晶片的效果。由此可知，光子處理器的運算系統可由較傳統的電子設備提供支援，執行部分 AI 任務時性能甚至優於最先進的全電子系統。

諸多自由空間光學處理器及運算、微軟類比運算機器



自由空間光學處理器

1. 大於 $\lambda/2$ ：由繞射表面厚度、折射率調變單元組成
2. 小於 $\lambda/2$ ：由金屬結構、介電單元組成
3. 這些自由空間處理器可以執行偏振處理、空間處理、通用線性變換以及波的光譜和時間處理。
4. Example: 2023 Microsoft's AIM (Analog Iterative Machine) 類比疊代處理器



光子量子計算 (Photonic Quantum Computing)

原理：

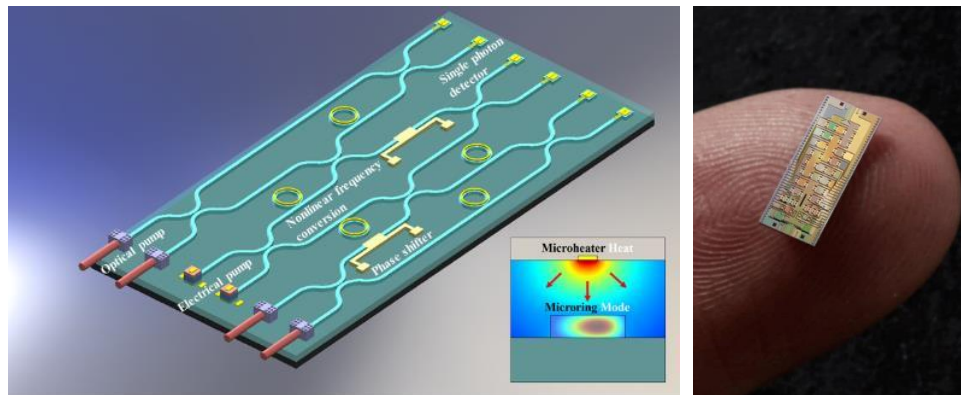
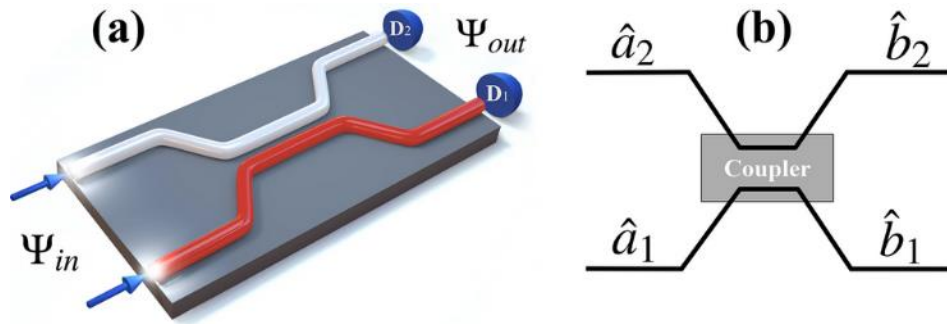
以光子的偏振、相位、時間或路徑作為 qubit，利用干涉與糾纏進行運算。

特點：

- 可在室溫運作，不需低溫。
- 易於量子通訊整合。

代表系統與廠商：

- *PsiQuantum* (U.S.) – 矽光子平台
- *Xanadu* (Canada) – Borealis 系統
- *Quantum Computing Inc*, QCI
- *Photonics Inc.*

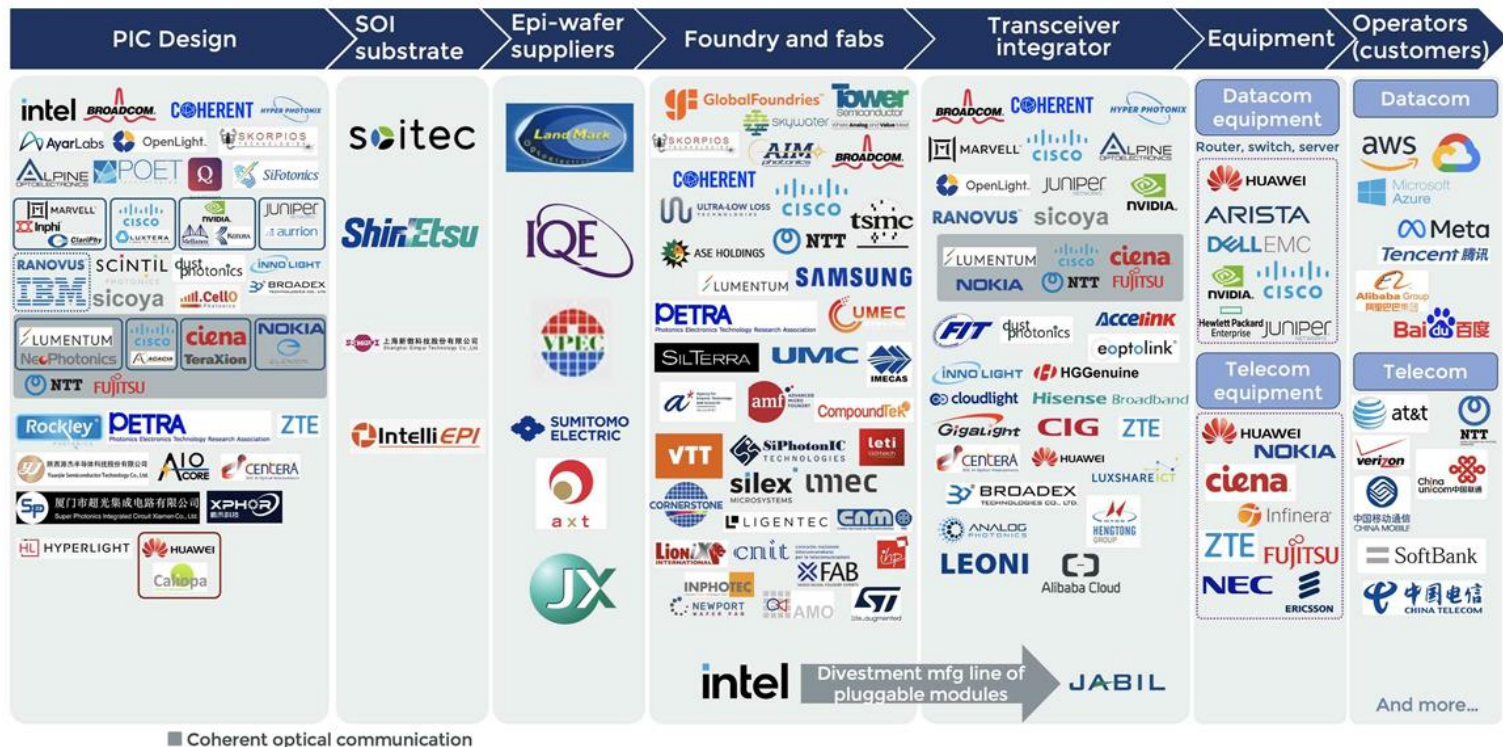


台灣光通訊產業生態系

產業層次	主要產品	主要廠商與機構	產業規模
光纖與光纜	光纖	卓越光纖	\$9億台幣
	光（電）纜	大亞電線*、太平洋電線*、台通光電*、華新麗華、聯合光纖等	~\$800
光通訊元件 （製造、代理）	光被動元件：光耦合器、光纖跳接線	上詮*、大地光纖、友嘉、光元先進、光合訊、光紅建聖*、光環*、光騰國際、佳必琪、昕鈺、東典光電、波若威*、前鼎*、索爾思、統新、逢源、創威*、崧銓、華信*、華星*、傳承、圓宏、寬原、歐鉉、聯合光纖、聯亞*、聯鈞*、鴻海*、曜隆等公司或部門	~\$2500
	光主動元件：雷射二極體、光收發模組、驅動IC		
光通訊設備、次系統、儀器、鋪設工程	光放大器、Switch、傳輸通信設備、電信寬頻產品、ONU、OLT、檢測設備	千才、台林通信*、台聯電訊*、仲琦*、合勤*、星通資訊*、亞旭電腦、捷耀、普萊德*、智邦*、等公司或部門	~\$3000
學術與研發機構	LED通訊、400G研究開發、教學	台大、工研院、中山、中華電信研究所、中興、北科大、台科大、交大、高應大、清大	

光通訊、半導體、化合物半導體、光學、光電、雷射

矽光子光通訊產業鏈



CPO 生態系：半導體、光學、封裝、設備與 AI 系統

- PIC Design (設計) : NVIDIA、Broadcom、Marvell、Ayar Labs、Lightmatter、Celestial AI

* 設計 PIC、Optical I/O。

- 雷射：Coherent、Lumentum

- SOI Wafer & Epi-wafer：Soitec、Shin-Etsu、SUMCO、GlobalWafers 等

* 提供 SOI、InP、GaAs 等光電材料。

- EIC / Retimer / SerDes / PHY：Marvell、Credo、MaxLinear、Semtech、Alphawave

* 高速 SerDes、DSP、Retimer、電光訊號轉換

- Connectors & Fibers (光纖互連)：Corning、US Conec、Molex、TE、YOFC

* MT/TMT、GlassBridge、光纖陣列、FAU、光纜

- ASIC / XPU Design (AI晶片)：NVIDIA、Broadcom、Intel、AMD、Marvell、Aster Labs

* GPU、Switch ASIC、AI Accelerator，是 CPO 的運算核心

• 晶圓代工

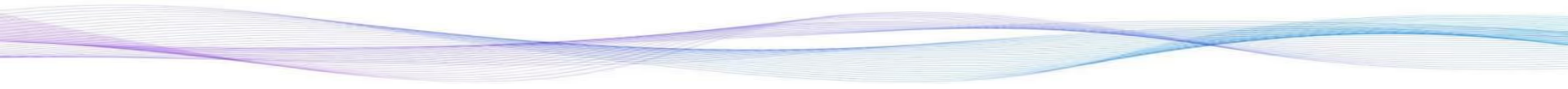
• 封裝 / 組裝 / 測試

• 系統整合商與雲端業者：負責將 CPO 導入機櫃級 AI 叢集與下一代資料中心架構。



- 雲端業者：Microsoft、Google、Meta、Amazon AWS
- ODM與硬體業者：緯穎、鴻海、廣達
- * 驗證 CPO 的系統效能、可靠度、維修。

The CPO Era: Restructuring of the Value Chain

- CPO technology shifts industry dominance from the traditional “pluggable optical module” to “EPIC + CPO”
 - The term – “Transceiver” will be replaced by “optical engine”.
 - As the value chain restructures, the roles of ASIC/GPU design giants and advanced packaging houses become increasingly crucial.
 - All manufacturers must migrate to the silicon photonics platform. Traditional production methods will gradually become obsolete.
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The words "The End" are written in a purple, cursive script font. The text is centered within a light blue, horizontally-oriented oval. Above and below the oval are decorative purple flourishes that resemble stylized floral or scrollwork patterns.

The End

感謝聆聽、敬請指教